Borrowed Ground: Land-sharing as insurance against drought in rural Tanzania

Kian Williams*

August 2026

Abstract

Droughts strike every household in a village at once, so the transfers rural communities use to insure one another dry up precisely when they are most needed. These covariate shocks, or shocks that affect the whole community simultaneously, are, on the standard view, uninsurable within the village. Using plot-level panel data from rural Tanzania, I show that drought shocks depress consumption-sharing but strengthen customary land-sharing. After a drought, reliance on "free-use" plots, borrowed at no charge from kin and neighbors, expands significantly, and the expansion is concentrated among the most vulnerable land-poor households. Lenders systematically give out peripheral, less valuable plots rather than prime ones, sacrificing relatively little of their own consumption. Land-sharing thus supplies a margin of insurance against the covariate shocks that defeat consumption-based risk-sharing, an insurance role empirically understudied in the literature.

*London School of Economics

1 Introduction

> *"In our traditional African society we were individuals within a community. We took care of the community, and the community took care of us... nobody starved, either of food or of human dignity, because he lacked personal wealth; he could depend on the wealth possessed by the community of which he was a member."* — Julius Nyerere, Tanzania's first President, quoted from *Ujamaa: The Basis of African Socialism.*
>
> *"When you reap the harvest of your land, you shall not reap to the very corners of your field... you shall leave them for the poor and the stranger."* — Leviticus 19:9–10, cf. the story of Ruth.

For many rural households, land is the foundation of economic life. This is acutely true in Tanzania, where more than 80% of the population relies on agriculture for its livelihood and 85–90% of cultivated land is worked by smallholders practicing rain-fed subsistence farming (FAO, a,b). Further, dependence on rain-fed agriculture leaves Tanzanian livelihoods exposed to growing climatic variability. Rising temperatures and longer dry spells leave Tanzania as the 26th most climate-vulnerable country in the world (USAID, 2018). For households whose welfare is bound to productive land, such shifts represent a direct threat to food security and livelihood.

In cases of idiosyncratic shocks to households (shocks specific to one household, such as an illness, a death, a lost job), a broad and established literature studies the evidence for risk-sharing, or better-off households supplementing consumption for households that have fallen on hard times (Cochrane, 1991; De Weerdt and Dercon, 2006; Mace, 1991; Townsend, 1994; Udry, 1994). The consensus is that partial risk-sharing occurs in many contexts, though this level of risk-sharing unsurprisingly falls short of the full-insurance benchmark, under which a household's consumption would move with village-wide resources and not with its own luck. Fafchamps and Lund (2003) and Fafchamps and Gubert (2007) show that these risk-sharing relationships tend to be more likely with greater geographic proximity and ties of formal kinship.

However, rainfall droughts constitute a *covariate* shock – meaning they affect most or all of the local community simultaneously. Townsend (1994) notes that covariate shocks leave every household short at once and little consumption left to share. Consequently, these shocks are deemed uninsurable within the village and assumed away. Kazianga and Udry (2006) supports this assumption empirically. Against the backdrop of a severe drought in Burkina Faso, the authors find little evidence of risk-sharing; however, their analysis limits the scope of risk-sharing to interhousehold transfers of consumable goods.[1]

[1] In Kazianga and Udry (2006), livestock and grain inventory are treated as self-insurance (own-asset smoothing), a conceptually separate category that is not part of their risk-sharing analysis.

In this thesis, I seek to understand whether risk-sharing can indeed occur through the sharing of land in the Tanzanian context. Relationship- and responsibility-based sharing of land is a significant part of customary tenure relationships that govern as much as 90% of land in Sub-Saharan Africa (Chimhowu, 2019). Why could land-sharing be any different from the consumption transfers or gifts that fail under covariate drought shocks? The answer lies in the particular nature of land as an asset. First, unlike consumable goods, land is a fixed, non-depletable asset that, if shared, can ultimately revert to its owner. Second, land is only productive to the extent that it is cultivated. Households face inherent constraints on labor and capital they can rent or themselves supply to cultivate plots. This is especially pertinent for land-rich households, who are more likely to possess peripheral plots that they leave uncultivated. In this case, a lender can extend land to others in need without surrendering their own consumption.[2] While the consumption-sharing margin has been studied extensively, gratis land-sharing as a response to covariate risk remains largely unexplored. It is this margin that this thesis takes up.

The above question yields three testable hypotheses. First, if free-use land is a risk-sharing arrangement, gratis arrangements should strengthen or even expand following drought shocks. Second, if land is unique because land-sharing bears low consumption costs to the lender, the plots given out should be their peripheral rather than prime plots. Third, if the arrangement shares risk, access should flow toward need; drought-induced borrowing should be concentrated among more vulnerable, land-poor households, with no comparable expansion among the land-rich.

Testing these hypotheses, I find that free-use plots are 14 percentage points *less* likely to revert to the owner following a drought season, compared to a baseline of 18% wave-to-wave. Beyond strengthening pre-existing land-sharing agreements, drought shocks further drive a significant *expansion* in holdings of free-use plots. Next, I turn to understanding the relative characteristics of freely-shared land as compared with other plots in a lender's portfolio. Lending households' given-out plots are roughly 19 percentage points less likely to have been cultivated in the prior season and are of lower value compared to the lender's portfolio, suggesting that these households are not sacrificing a significant portion of their own immediate consumption and productivity in lending them. Finally, the borrowing is driven by land-poor households, suggesting a flow of land based on vulnerability. These results are robust to different rainfall shock measures, sample restrictions, and a battery of specifications. Taken together, my findings provide evidence that the sharing of plots of land, largely understudied in the economics literature yet commonplace in many parts of

[2]Like many forms of insurance (though unlike immediate consumption-sharing), land lending usually represents a delayed response to the shock. Though the welfare consequences of the growing season shock materialize in the year following the harvest $(t+1)$, gratis land provides welfare most directly after the subsequent harvest $(t+2)$. The benefits of land-sharing in improving recipient welfare and replenishing their grain stores and savings are therefore offset by one agricultural year.

sub-Saharan Africa, plays a role in buffering rainfall shocks for the most vulnerable rural households when other forms of community risk-sharing fail.

2 Literature and Contribution

This thesis contributes to three main bodies of literature. The first studies the mechanisms of and requirements for community risk-sharing (Cochrane, 1991; Kazianga and Udry, 2006; Mace, 1991; Townsend, 1994, 1995; Udry, 1994). As discussed in Section 1, the consensus is that covariate shocks are uninsurable through risk-sharing. This thesis extends the scope of risk-sharing to land across which covariate drought risk can indeed be shared.

Second, an established literature measures the economic gains from expanding the priced margin of land access.[3] This body of work finds that reforms that marketize land shift ownership towards more productive farmers and have positive impacts on output (Acampora et al., 2025; Beg, 2022; Chari et al., 2021; Chen, 2017; de Janvry et al., 2015; Gottlieb and Grobovšek, 2019; Manysheva, 2022). Chen et al. (2023) frames the premise: in Malawi, 84.5% of farm households operate only non-marketed land,[4] and an efficient allocation of land would raise output by a factor of 1.7 to 2.8. The unpriced (inclusive of free-use) margin is quantified and framed as a misallocation to be removed. My paper emphasizes the insurance role of non-marketed land during shocks in rural Tanzania, a function that is invisible in productivity accounting.

A third parallel, largely qualitative literature describes social norms around customary land allocation in Sub-Saharan Africa. It holds that land is often an entitlement embedded in social relations rather than simply a productive asset, and that land loaned or granted without payment is pervasive (Berry, 1993; Lavigne Delville et al., 2002; Platteau, 2000). These observations ground the dynamics that I study in this paper. Platteau (1996), for example, documents customary land's insurance role: to the extent that land arrangements continue "to fulfill an insurance function... the risk of social exclusion and economic pauperization reflected in landlessness... is thus averted" (p. 52). Platteau then posits that land's role as a productive asset, highlighted in the priced-margin literature, is subsidiary to its role as social insurance: "[In] African village communities... considerations of social security and equity usually dominate considerations of pure efficiency" (p. 75). This literature remains largely observational, and this claim is never studied empirically. My paper empirically tests these findings, demonstrating that unpriced land access in Tanzania expands for vulnerable households precisely when drought raises their need.

More broadly, this paper also speaks to the growing body of work that studies how

[3]Note Gollin and Udry (2021) argue these reallocation gains are overstated once measurement error and unobserved heterogeneity are accounted for.

[4]For example, plots acquired by inheritance or bride price, shared, or chiefly allocated.

households cope with negative income shocks. These coping mechanisms include migration (Meghir et al., 2022; Morten, 2019) or asset sales (Janzen and Carter, 2019). Other papers specifically study the role of land in coping with shocks. Eskander and Barbier (2023), Tione and Holden (2021), and Kusunose and Lybbert (2014) study how households use transactions on the land rental market to buffer weather shocks. Veljanoska (2018) studies how spatially fragmented land portfolios help improve farmers' resistance to climate shocks. Finally, specific to the Tanzanian context, a related paper Murken et al. (2024) identifies the role of rainfall shocks in contributing to land conflicts and incentivizing households to seek formal land certification. Where land does enter the *empirical* analysis of shocks, it forms a channel of adjustment via the *priced* market (sold, leveraged for credit, rented in or out). This paper studies a novel dynamic: the gratis lending and borrowing of land between households ("free-use" land) as a mechanism of coping. I am aware of no study that takes up this margin of response to covariate shocks.

3 Data

3.1 The Tanzania National Panel Survey

The household data used in this paper come from the Tanzania National Panel Survey (NPS), a nationally representative survey implemented by the Tanzania National Bureau of Statistics. A fixed sample of households is re-interviewed, and split-off individuals who form new households are tracked and re-surveyed, allowing households to be linked over time. Alongside standard modules on demographics, consumption, and assets, each round administers a detailed agricultural questionnaire at the plot-level, the core of this study.

I use the first three waves of the NPS, fielded in 2008–09, 2010–11, and 2012–13: the roughly 3,300 households surveyed across 409 enumeration areas at baseline are tracked forward through the later two rounds and matched using the survey's official panel key.[5] The resulting dataset is a three-wave, plot-level panel in which each plot is nested within households.

In the NPS, a plot is a contiguous piece of agricultural land under a single tenure arrangement, described by the household member managing it. Across the three waves, the survey documents on the order of five to nine thousand plots per round, giving the panel substantial within-household, cross-plot variation.

Finally, each NPS round fields separate general household and community modules administered to local leaders. I draw on these where they speak to community- and household-level risk-sharing. The household modules provide information on gifts and transfers that

[5] I restrict attention to these waves because they are the only stretch of the NPS over which households can be followed continuously.

households receive. The community-level data are matched to households geographically through their shared enumeration area (EA) and capture community-level features such as participation in farmer cooperatives.

Plot tenure and plot loss Two plot-level attributes are central to my analysis. The first is *ownership status*: each plot is classified as owned, used free of charge, rented in, or sharecropped. For the analysis, I collapse tenure status into three groups. *Owned* covers plots held outright. *Free-use* plots are borrowed gratis from a specific individual. *Rented* pools cash-rented plots with sharecropped-in plots, since both are arrangements in which the household works land it does not own and pays the owner in cash or output. Owned plots dominate at roughly 83–84% in every wave, with borrowed free-use plots making up around 11–12% and rented-in plots a smaller 3–5%. The three tenure types vary considerably across observable characteristics. Appendix Table 1 shows owned plots are much larger (about 3 acres, versus 1.3 for free-use and 1.7 for rented) and closer to the dwelling, while rented plots lie furthest away. A joint t-test rejects equality of the three groups for every characteristic.

Free-use borrowing is the focus of my thesis. These arrangements are overwhelmingly local and kin-based. Of free-use plots with an identified owner, relatives are by far the largest source (about 45%), followed by neighbors (about 19%) and a residual of friends, employers, and other local contacts. This pattern is consistent with free-use land operating as need-based reallocation within the kin network rather than an impersonal market (Berry, 1993; Platteau, 2000).

The second attribute is *plot retention and loss*. Working from a lineage-level plot panel,[6] a plot is coded as lost if it was on the household's roster in the prior wave but absent in the current one, conditional on the lineage being tracked across both waves. By construction, the plot is otherwise retained. Appendix Figure 1 maps plot loss by wave across Tanzania.

Agricultural productivity I measure agricultural productivity at the plot–year level with three variables, all drawn from the post-harvest plot roster. *Harvest value* is the imputed real value of the plot's harvest, where physical quantities are valued at median wave–enumeration-area crop prices and deflated to base-2010 Tanzanian shillings. *Harvest value per acre* is the same imputed real harvest value taken per acre harvested. *Maize yield* is the physical quantity of maize harvested per acre harvested, in kilograms per acre. Maize yield is included because the two value-based measures rely on imputed prices, so their cross-wave variation may in part reflect rainfall-driven movements in local crop prices rather than physical output. Maize is a dominant staple grown by the majority of rural Tanzanian

[6] A *household-lineage* is the dynasty descending from an original wave 1 household. The plot-level analysis is conducted at this level so that plots can be followed across household fission, for example, an adult family member leaving the household but retaining ownership of a plot from the previous household.

households, making it a widely observed welfare-relevant measure of productivity.

All three outcomes enter in logarithms. Output and harvest value are transformed as $\ln(1+x)$ so that plot–years with a zero-value harvest are retained. The log transformation is for two reasons. First, the value measures are extremely right-skewed (Appendix Figure 2): a small number of plot-years with very little harvested area generate per-acre values several orders of magnitude above the median. Second, log outcomes give the shock coefficient a semi-elasticity interpretation (approximately the percent change in output per one-unit increase in the drought index) which is comparable across the three outcomes despite their different units.

3.2 Rainfall shocks

I measure rainfall using CHIRPS, daily gridded precipitation data at 0.05° (≈5.5 km) resolution that blends satellite estimates with station data. CHIRPS gives consistent coverage across all enumeration areas (EAs). I extract the daily series at each household's enumeration-area coordinates.

For the measure itself, I use the Standardized Precipitation Index (SPI) of McKee et al. (1993), well-regarded for measuring rainfall deficit and surplus. For each location I accumulate rainfall over a fixed window, fit a gamma distribution to a 1981–2007 calibration period, and map each survey year's accumulation to a standard-normal deviate. The SPI is well suited to this paper: it is location-specific, so a value of -1 means "one standard deviation below the local norm" and is comparable across the panel's heterogeneous agro-climatic zones; the gamma fit respects the non-negative nature of rainfall; and calibrating on 1981–2007, which predates the survey, keeps the reference benchmark uncontaminated by the years used in estimation.

My primary measure, SPI-6, accumulates rainfall over the six months ending in May (December–May), capturing the total water supply over the main growing season. As a robustness check I also use SPI-3, accumulated over March–May, the core of the long rains. The motivation follows Fishman (2011) and Dell et al. (2014): the distribution of rain across the season matters as much as the total. Conditional on the same seasonal total, a poorly timed season can still depress yields. Because the two windows aggregate differently, SPI-6 reflects overall supply while SPI-3 weights the heart of the season.

By convention, negative SPI denotes dry conditions. However, because the object of interest is dry periods, I flip the sign of all rainfall variables so that *higher values denote drier conditions*. A positive coefficient therefore means drought raises the outcome.

For each household I compute the index over the prior season for each wave. Because the NPS coordinates are enumeration-area-level, the SPI is a fixed enumeration-area quantity. I can cleanly merge it onto the panel with no imputation. Appendix Figure 3, Appendix

Figure 4, and Appendix Figure 5 map drought shocks plotted by wave.

4 Methodology

4.1 Empirical strategy

The identifying premise of this thesis is that conditional on location, the dry spell a household faces in a given season is plausibly random. This property has made weather a workhorse source of exogenous variation (Dell et al., 2014; Miguel et al., 2004).

I exploit this variation within a panel fixed-effects design. The core household-level specification is

$$y_{ht} \;=\; \beta\,\text{Drought}_{e(h),\,t-1} \;+\; \mathbf{X}'_{ht}\gamma \;+\; \alpha_h \;+\; \theta_t \;+\; \varepsilon_{ht}, \tag{1}$$

where y_{ht} is an outcome for household h in wave t – primarily its reliance on free-use land measured as a share of the plot portfolio and as a count; $Drought_{e(h),\,t-1}$ is the rainfall shock in the enumeration area $e(h)$ in which the household sits, lagged one agricultural year; α_h is a household fixed effect and θ_t a wave fixed effect. The plot-level specification is the analogue,

$$y_{it} \;=\; \beta\,\text{Drought}_{e(i),\,t-1} \;+\; \mathbf{X}'_{it}\gamma \;+\; \alpha_{\ell(i)} \;+\; \theta_t \;+\; \varepsilon_{it}, \tag{2}$$

in which the unit is the plot i and $\alpha_{\ell(i)}$ is a fixed effect for the household-lineage $\ell(i)$ to which the plot belongs, so that plots are compared within the same lineage across waves (Section 4.2). β is the parameter of interest: the within-unit response of the outcome to a deterioration in rainfall.

The household (or plot-lineage) fixed effect α_h absorbs every time-invariant feature of the unit so that β is identified only from *changes* in the outcome within the same household as the shock varies over time. The wave effect θ_t removes anything common to all households in a given round – national price movements, survey-wide changes in fieldwork, and aggregate trends – so that the relationship is identified from idiosyncratic local shocks rather than from the national time series.[7] This design follows closely that of Murken et al. (2024), who study rainfall shocks and land outcomes in the same survey with an analogous within-household estimator.

Standard errors are clustered at the EA throughout, following Murken et al. (2024). Because a rainfall shock is common to households in the same locality, their outcomes are not independent within an EA, and this is the natural level at which to allow for this correlation. Note that I confirm the results are robust to other forms of spatial clustering (Section 4.2).

[7] As Dell et al. (2014) put it, "[t]ime-fixed effects θ_{rt} further neutralize any common trends and thus help ensure that the relationships of interest are identified from idiosyncratic local shocks."

Controls I deliberately keep the vector of time-varying controls $\mathbf{X}$ sparse. Given that household and wave effects absorb fixed heterogeneity and common shocks, additional regressors aim to address residual variation.[8] I therefore restrict $\mathbf{X}$ to credibly exogenous controls – household head age, gender, and highest grade of education completed,[9] and treat the weather variation net of the two sets of fixed effects as the source of identification rather than introducing covariates that are themselves plausibly downstream of the shock.

The lagged shock The shock enters with a one-year lag because the outcomes are land-portfolio decisions taken under the conditions the preceding agricultural year left behind. Tanzania's dominant crop is planted around March and harvested May–August, while each NPS wave is fielded over a full year, so most interviews (roughly 56%, near 60% by wave 3) occur before the current harvest begins.[10] For these households, the scarcity they face – grain stores being drawn down, land arrangements being entered or exited – is conditioned by the *prior* year's harvest. Appendix Table 17 and Appendix Table 18 confirm the results are directionally unchanged when the sample is restricted to pre-harvest interviews, for whom the timing argument holds unambiguously.

Household versus plot fixed effects Equation (2) uses a household-lineage rather than a plot fixed effect for a structural reason specific to the loss outcome. The plot-loss indicator is defined only in a plot's *loss-eligible* waves, or those in which the household held the plot in the previous wave. Therefore, a plot contributes at most two observations: one at wave 2 (if held in wave 1) and one at wave 3 (if held in wave 2). A plot fixed effect compares a plot only against itself and is therefore identified solely from plots observed in *both* transitions, that is, plots retained through wave 2. This has two consequences: every plot lost in the first (wave 1 to wave 2) transition has a single loss-eligible observation and is dropped, discarding roughly half of all loss events; and identification collapses onto the second transition among the plots that survived to wave 2, conditioning the estimate on survival through the first transition. The household-lineage effect is the tightest comparison that avoids this. It holds fixed the household that holds the plot while still permitting plots to enter and exit, and early losses contribute to the coefficient. Where the outcome *is* defined on continuously-observed plots

[8] Dell et al. (2014) state the principle directly: "to the extent that the Z_{it} are endogenous to the weather variation, the 'over-controlling' problem that complicates cross-sectional estimation appears in the panel context as well... Best practices suggest including only credibly exogenous regressors as control variables Z_{it} ... Potentially endogenous regressors should typically only be included if there is a strong argument that these variables are not affected by climate."

[9] As discussed in Section 4.2, Appendix Table 5 shows that the household head, and thus their characteristics, does not change as a result of the treatment.

[10] These numbers are conservative. The harvest period begins in May, but many households continue to bring in crop until the end of the summer. Further, any bias from a successful concurrent harvest immediately prior to the interview would likely be attenuating.

– notably the productivity measures, which are recorded in every wave a plot is cultivated – I additionally report plot fixed-effects specifications (Appendix Table 6), which absorb time-invariant plot quality directly and serve as a robustness check on the primary results.

The geographic level of the shock Following Murken et al. (2024), rainfall is measured at the EA rather than at the individual plot. This is appropriate for two reasons. First, a covariate dry spell strikes a whole locality at once, and the enumeration area is a unit within which rainfall is approximately common. Second, the spatial resolution of the gridded rainfall data is coarser than the within-EA spread of plots, so plot-specific assignment would not add any meaningful precision.

4.2 Addressing threats to identification

Attrition Throughout, the household-level analysis is restricted to rural households and the plot-level analysis to rural plots held by household-lineages (described in note 6) that are rural in every wave. Both analyses use the full, unbalanced panel rather than discarding households or lineages seen in only two of the three waves, which would sacrifice power and condition on survival.

This within-unit fixed-effects estimator is unbiased on an unbalanced panel under the condition that attrition is unrelated to both the shock and the change in outcome. Therefore, I confirm that differential attrition does not drive the results. First, drought shocks do not raise household or plot-lineage dropout (Appendix Table 2). Second, although in the household sample attritors differ from stayers on demographics, they do not differ on the free-use share or the shock; in the plot-lineage sample, retained and dropped lineages differ on neither the shock nor any portfolio variable (Appendix Table 3; the household-frame differences in plot counts are marginal and, as level differences, are absorbed by the fixed effects). Third, neither the within-household change in free-use reliance nor a lineage's free-use holding, the variation identifying the fixed-effects estimates, predicts attrition (Appendix Table 4). Finally, I show that the results are consistent with a balanced-panel restriction. This specification retains only households (or household-lineages in the plot-level analysis) tracked in every wave, and leaves both the plot-loss results (Appendix Table 10) and the household-level results (Appendix Table 14) essentially unchanged.

Household compositional change A second concern is that the within-household estimator attributes a change in free-use reliance to the same household responding to drought, which presumes the household is a stable unit. If instead drought changes who the household is, say, through the death or replacement of the head, or causes it to lose members, the within-household change could reflect a shift in composition rather than behavior. Appendix Table 5 shows it does not. Regressing wave-to-wave changes in household composition on

the shock, drought predicts neither a change in the sex of the head, nor a head replacement, nor a decline in household size, and the free-use results are not driven by lineage splits (Appendix Table 16). The household is therefore stable with respect to the shock, so the results reflect behavior rather than compositional change.

Clustering A further concern is that EA clustered standard errors may understate uncertainty if rainfall shocks and land access outcomes are spatially correlated across EA boundaries, since a covariate shock by its nature strikes contiguous areas simultaneously. To address this I confirm that results hold using Conley spatial standard errors (Conley, 1999) via the `acreg` estimator (Colella et al., 2019), which allow for spatial correlation between any two units within a chosen distance cutoff. The main results are robust; both the plot loss (Appendix Table 13) and the household free-use reliance estimates (Appendix Table 15) retain their significance at spatial cutoffs of 50, 100, and 200 km.

5 Results

5.1 First-order impact of drought shocks on agricultural productivity

I begin by testing the first-order impact of drought shocks (as defined through the SPI-6 measure discussed in Section 3.2) on agricultural productivity (specific measures discussed in Section 3.1). To do this, I build from Equation 2.[11] The results are reported in Table 1 below. I find that a growing season with one standard deviation less rainfall than its 1981–2007 local norm lowers a plot's harvest value per acre by about 11% ($\beta = -0.119$, s.e. $= 0.041$; significant at the 1% level), its total real harvest value by about 8% ($\beta = -0.079$, s.e. $= 0.039$; significant at the 5% level), and its maize yield by about 7% ($\beta = -0.069$, s.e. $= 0.041$; significant at the 10% level). All results are robust to the replacement of household-lineage fixed effects with plot fixed effects (Appendix Table 6).

Confirming this impact has two purposes. First, it provides validation as to the materiality of the shock measure – drought shocks as defined have an expected impact on agricultural households. Second, it confirms the need for risk-sharing that underlies the remainder of the paper. Because drought shocks depress the productivity and output of the agricultural land households rely on, they indeed constitute a covariate shock to household welfare.

[11] Note that because the outcome is the harvest itself, the shock here instead enters *contemporaneously*, and the sample is restricted to interviews conducted after the concurrent harvest period begins.

Table 1: Rainfall shocks and agricultural productivity

	Log harvest value (p.a.) (1)	Log harvest value (2)	Log maize yield (3)
Drought shock (concurrent)	-0.1189***	-0.0794**	-0.0685*
	(0.0408)	(0.0390)	(0.0407)
Mean of dep. var.	11.378	11.679	5.555
SD of dep. var.	1.087	1.354	0.962
Wave fixed effects	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes
HH fixed effects	Yes	Yes	Yes
Observations	3,176	3,179	2,150
Households	687	687	562
Clusters	119	119	113
Within R^2	0.014	0.011	0.006

Notes: The unit of observation is a plot-year. The sample is the plot-year panel from the World Bank LSMS Tanzania National Panel Survey waves 1–3 (2008/09–2012/13), restricted to household-lineages rural in every wave and to plots interviewed after the concurrent long-rains harvest. Each column is a separate within-lineage fixed-effects regression: (1) log real harvest value per acre (base-2010 TSH), (2) log real harvest value (base-2010 TSH), (3) log maize yield. **Drought shock (concurrent)** is the six-month Standardized Precipitation Index for May of the survey year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. All specifications absorb household-lineage fixed effects and include wave fixed effects and time-varying controls for head sex, head age, and head education (highest grade completed). Households is the number of distinct lineages contributing to the estimate. Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.2 Drought shock impacts on consumption-sharing and transfers

I now show that drought shocks are associated with a breakdown in traditional consumption-sharing between households (cash transfers, gifts), presented in Table 2. This depression in consumption-sharing occurs when it is reasonable to expect the need for cash and inter-household consumption-sharing to peak.

Table 2: Rainfall shocks and household risk-coping transfers

	Cash assistance	Relied on friend	Days relied	Any food gift	No. food gifts
	(1)	(2)	(3)	(4)	(5)
Drought shock (prior)	-4348.031*	-0.026**	-0.088**	-0.010	-0.006
	(2623.295)	(0.013)	(0.040)	(0.014)	(0.027)
Mean of dep. var.	3537.905	0.087	0.207	0.447	0.811
SD of dep. var.	92914.542	0.281	0.821	0.497	1.226
Wave fixed effects	Yes	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes	Yes
HH fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	4,353	3,014	3,014	4,353	4,353
Households	2,022	1,862	1,862	2,022	2,022
Clusters	327	325	325	327	327
Within R^2	0.005	0.005	0.006	0.008	0.008

Notes: The sample is rural households in the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13). Each column is a separate within-household fixed-effects regression of the column outcome on the rainfall shock. Outcomes are: (1) total cash assistance received over the past 12 months (TSH); (2) an indicator for having relied on a friend or relative for food in the last 7 days; (3) the number of days in the last 7 the household relied on a friend or relative for food; (4) an indicator for having consumed any food received as a gift in the last 7 days; and (5) the number of food items received as a gift in the last 7 days. **Drought shock (prior)** is the six-month Standardized Precipitation Index (SPI-6) for May of the prior year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. All specifications include household fixed effects, wave fixed effects, and time-varying household controls for head sex, head age, and head education (highest grade completed). Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Consistent with Kazianga and Udry (2006), I find that growing season drought shocks lead to reduced received cash assistance from other households over the prior 12 months ($\beta = -4348.031$, s.e. $= 2623.295$; significant at the 10% level and large compared to a sample mean of ≈ 3500 TSH), a 2.6 percentage point decrease in the probability of having relied on a friend for food in the past 7 days ($\beta = -0.026$, s.e. $= 0.013$; significant at the 5% level and an $\approx 30\%$ reduction compared to a sample mean of 8.7%), and a decrease in the number of days households relied on a friend for food in the past 7 days ($\beta = -0.088$, s.e. $= 0.040$; significant at the 5% level though modest in magnitude). Though not precisely estimated, directionally similar results appear for whether households received any food *gift*

over the past 7 days ($\beta = -0.010$, s.e. $= 0.014$; modest effect size) and the number of food gifts received ($\beta = -0.006$, s.e. $= 0.027$; modest effect size).

Appendix Table 7 shows that this breakdown in consumption-sharing extends beyond inter-household transactions and has implications for community activities and participation. Shocks drive a decline in the number of farmer cooperatives ($\beta = -1.582$, s.e. $= 0.798$; significant at the 5% level). These farmer cooperatives are member-owned organizations where smallholders pool their resources to improve crop marketing, access farming supplies, and secure better prices. Though not significant at conventional levels, directionally similar results appear for related outcomes: the number of cooperative-associated members ($\beta = -31.005$, s.e. $= 23.852$), funds raised for village projects ($\beta \approx -3,892,000$, s.e. $\approx 3,324,000$), and the number of village projects undertaken ($\beta = -0.093$, s.e. $= 0.091$).

As predicted in the work of Townsend (1994), Kazianga and Udry (2006), and others, drought (covariate) shocks indeed lead to a depression in household risk-sharing in my sample. Transfers and gifts between households decrease as sharing consumption becomes more costly for all. I also identify a consistent corollary story at the village-level: shocks depress community engagement, particularly engagement that requires upfront resource commitment or fees such as farmer cooperatives and village project investments.

5.3 Drought shock impacts on land-sharing

Drought shocks strengthen free-use plot retention Having established that the covariate shock lowers inter-household transfers, I now turn to the land margin itself and ask how the same shock shifts the arrangements through which households hold land. As in Equation 2, the dependent variable is an indicator equal to one if a plot held by the household-lineage in wave $t-1$ has left its portfolio by wave t, and I estimate the relationship on the subsample of plots that were free-use in the prior wave, absorbing household-lineage fixed effects throughout so that the comparison is between plots of the same lineage across waves. The results are presented in Table 3. Among free-use plots, a drought shock drives a large 14.3 percentage point *decrease* in the probability of loss ($\beta = -0.143$, s.e. $= 0.057$; significant at the 5% level), a $\approx 78\%$ reduction on the 18.3% base rate.[12]

[12]The free-use finding is robust to limiting to plots lent by a close connection (kin, friend, employer, or neighbor), reinforcing the interpretation that this mechanism runs through land-sharing along within-community social ties (Appendix Table 8).

Table 3: Rainfall shocks and loss of free-use plots

	Dependent variable: plot lost (=1)
	Free-use
	(1)
Drought shock (prior)	-0.1432**
	(0.0572)
Mean of dep. var.	0.183
SD of dep. var.	0.387
Wave fixed effects	Yes
Demographic controls	Yes
HH fixed effects	Yes
Observations	378
Households	129
Clusters	84
Within R^2	0.023

Notes: The unit of observation is a plot-year. The dependent variable is an indicator equal to one if a plot held by the household-lineage in wave $t-1$ is absent from its portfolio in wave t. The sample is the plot-year panel from the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13), restricted to household-lineages that are rural in every wave. The sample is limited to free-use plots: a plot enters the estimation sample only if it was used free of charge (borrowed) in the previous wave. **Drought shock (prior)** is the six-month Standardized Precipitation Index (SPI-6) for May of the prior year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. The specification absorbs household-lineage fixed effects and includes wave fixed effects and time-varying household controls for head sex, head age, and head education (highest grade completed). Households is the number of distinct lineages contributing to the estimate. Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

To highlight the uniqueness of this result, Appendix Table 9 presents the same regression estimated across the full sample, and the owned, rented, and free-use tenure subsamples. In the pooled sample, growing season drought shocks bear essentially no statistically significant relationship to plot loss ($\beta = 0.015$, s.e. $= 0.012$), and the same is true of the owned plots that make up the overwhelming majority of the panel ($\beta = 0.018$, s.e. $= 0.012$). Owned land is likely not the margin on which drought operates: households do not sell or surrender the plots they own outright. However, the disaggregation by tenure reveals two effects

of opposite sign. Among plots rented in or held under sharecropping, a drought shock is associated with a 24.1 percentage point increase in the probability that the plot leaves the household's portfolio ($\beta = 0.242$, s.e. $= 0.066$; significant at the 1% level); against a base loss rate of 48.3%, a shock of one standard deviation raises the loss probability by half its base rate. Among free-use plots, by contrast, the same shock drives a 14.3 percentage point *decrease* in the probability of loss, the large effect described above.

In other words, the effect found on free-use plots does not simply capture a broad scramble for arable land across tenure types during shocks. Drought shocks do not uniformly cause households to enter or exit land arrangements. They have no measurable effect on owned land. They sever the priced rental contract, formally contingent on ability-to-pay. And they strengthen free land-sharing, the hypothesized margin of risk-sharing. These results are robust to limiting to the balanced panel (Appendix Table 10), to censoring wet years to address possible flooding damage to crops and soil implicitly captured by SPI-6 (Appendix Table 11),[13] to the replacement of the shock measure SPI-6 with SPI-3 (Appendix Table 12, discussed in Section 3.2), and to the replacement of EA standard error clustering with Conley spatial standard errors (Appendix Table 13).

The tenure contrast in Table 3 rests on a separately estimated subsample, which leaves open whether the free-use and non-free-use slopes are in fact statistically distinguishable from one another. Table 4 addresses this by pooling the panel and interacting the shock with an indicator for the plot having been used free of charge in the prior wave, absorbing household-lineage-by-tenure fixed effects so that each tenure group carries its own lineage intercept. The differential survives. Under the six-month drought shock, the interaction term is -0.163 (s.e. $= 0.056$; significant at the 1% level) against a non-free-use main effect of 0.020 (s.e. $= 0.012$; significant at the 10% level), implying a free-use slope matching Table 3 of roughly -0.143: a rainfall deficit shock that leaves the loss probability of the rest of the portfolio essentially unmoved reduces the loss probability of borrowed plots. A similar pattern holds under the three-month index. Therefore, the response of borrowed land to a drought shock is significantly different in sign and magnitude from that of the land a household holds by any other arrangement.

[13] The primary reading of SPI-6 treats higher values as worse growing conditions for Tanzanian farmers. However, very low values of SPI-6 indicate very wet years that damage crops. Appendix Table 11 addresses this by imposing a "floor" of 1 standard deviation wetter than usual on the SPI-6, so that identification comes from variation on the normal-to-dry range. The coefficients on free-use and rental plots are robust, and that on owned plots is positive and becomes statistically significant, entirely consistent with the headline findings.

Table 4: Rainfall shocks and free-use plot loss (interaction)

	Dependent variable: plot lost (=1)	
	Drought shock (1)	Drought shock (3-mo) (2)
Drought shock (prior)	0.0203*	0.0226**
	(0.0118)	(0.0099)
Drought shock × Free-use	-0.1634***	-0.1573***
	(0.0563)	(0.0408)
Mean of dep. var.	0.113	0.113
SD of dep. var.	0.316	0.316
Wave × tenure fixed effects	Yes	Yes
Demographic controls	Yes	Yes
HH × tenure fixed effects	Yes	Yes
Observations	5,439	5,439
Households	1,153	1,153
Clusters	234	234
Within R^2	0.005	0.006

Notes: The unit of observation is a plot-year. The dependent variable is an indicator equal to one if a plot held by the household-lineage in wave $t-1$ is absent from its portfolio in wave t. The sample is the plot-year panel from the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13), restricted to household-lineages that are rural in every wave. Each column interacts the rainfall shock with an indicator for the plot having been used free of charge (borrowed) in the previous wave; column (1) uses SPI-6 and column (2) uses SPI-3. **Drought shock (prior)** and **Drought shock (3-mo, prior)** are the six- and three-month Standardized Precipitation Indices for May of the prior year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. All specifications absorb household-lineage-by-prior-tenure fixed effects (saturated FEs) and wave-by-prior-tenure fixed effects, which allow each tenure group its own wave profile of plot loss; they include time-varying household controls for head sex, head age, and head education (highest grade completed). Households is the number of distinct lineages contributing to the estimate. Standard errors clustered at the enumeration area in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Drought shocks expand borrowing of free-use plots The previous section shows that drought periods protect pre-existing free-use lending arrangements: free-use plots are significantly less likely to be lost from the borrower's portfolio following the shock. It remains to show that drought also *expands* free-use borrowing. To test this, I investigate how shocks affect the number of free-use plots that households hold.

Applying Equation 1, Table 5 presents the impact of drought shocks on the share, number, and presence of free-use plots in a household's land portfolio. All three margins move upward are are precisely estimated. A growing season drought shock is associated with a 1.25 percentage point increase in the share of a household's plots held free of charge ($\beta = 0.013$, s.e. $= 0.005$; significant at the 1% level), which against a mean free-use share of 10.9% amounts to a shift of roughly a tenth of baseline reliance. The extensive and intensive

margins tell the same story: the shock is associated with an increase of 0.027 in the count of free-use plots a household holds ($\beta = 0.027$, s.e. $= 0.014$; significant at the 5% level, against a mean of 0.240 plots), and with a 1.76 percentage point increase in the probability that a household holds any free-use plot at all ($\beta = 0.018$, s.e. $= 0.009$; significant at the 5% level, against a base rate of 25.2%). The effects are modest in absolute size (a household does not restructure its portfolio wholesale in response to one bad season) but they are consistent across every way of measuring free-use reliance, and the consistency is informative: the share result is not being driven mechanically by households shedding other plots, since the count and the incidence of free-use holdings rise in levels alongside it. Therefore, drought shocks drive not just a reduction in the probability of losing a free-use plot (Table 3), but also an increase in the share, presence, and simple count of free-use plots held by households.

Table 5: Rainfall shocks and household reliance on free-use land

	Share free-use	No. free-use	Any free-use
	(1)	(2)	(3)
Drought shock (prior)	0.0125***	0.0272**	0.0176**
	(0.0048)	(0.0137)	(0.0086)
Mean of dep. var.	0.109	0.240	0.252
SD of dep. var.	0.276	0.640	0.434
Wave fixed effects	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes
HH fixed effects	Yes	Yes	Yes
Observations	3,874	3,874	4,353
Households	1,748	1,748	2,022
Clusters	295	295	327
Within R^2	0.006	0.005	0.004

Notes: The unit of observation is a household-year. The sample is rural households in the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13). Each column is a separate within-household fixed-effects regression of the column outcome on the rainfall shock. Outcomes are the share of the household's plots that are used free of charge (borrowed) in column (1), the number of free-use plots in column (2), and an indicator for holding any free-use plot in column (3). **Drought shock (prior)** is the six-month Standardized Precipitation Index (SPI-6) for May of the prior year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. All specifications include household fixed effects, wave fixed effects, and time-varying household controls for head sex, head age, and head education (highest grade completed). Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The headline pattern is clear. My findings parallel the observation of Kazianga and Udry (2006) and Townsend (1994) that consumption-based risk-sharing fails under covariate

shocks, but the unexplored land margin tells a different story. That is, drought shocks lead to an increase in free-use land in household portfolios, consistent with a gratis reallocation of land between households.

Lenders share peripheral plots in their portfolios Now I study the relative role that free-use (or freely given-out) land plays in the portfolio of its holder. As discussed in Section 1, one reason why land can theoretically remain a risk-sharing margin when consumption-sharing is costly lies in the nature of land as a productive asset only to the extent that it is cultivated. Therefore, one could expect lenders to lend land that they themselves were unable to cultivate, or that is of relatively lower-quality compared to the rest of their land such that the opportunity cost of lending is relatively low. Under these conditions, lending these plots constitutes little sacrifice in consumption to lenders when scarcity imposed by rainfall shocks makes consumption-sharing costly.

To do this, I compare free-use plots to the rest of a household's portfolio on both the lender and borrower side. On the lender side, the NPS data permits me to identify owned plots that a household has given out without charge. On the borrower side, I identify free-use plots as before. The findings are strong and consistent.

On the lender-side, given-out plots are relatively worse, peripheral plots in their portfolios. The results are presented in Table 6. First, plots given-out in the current season are 19 percentage points less likely to have been cultivated by the owner in the previous wave ($\beta = -0.186$, s.e. $= 0.085$; significant at the 5% level). Relatedly, they are ≈ 18 percentage points more likely to be left fallow by the owner in the previous wave ($\beta = 0.184$, s.e. $= 0.065$; significant at the 1% level). This is especially notable when compared to a baseline mean fallowing rate across all plots of 12%. Given-out plots are also assigned a lower resale value relative to the rest of the portfolio ($\beta = -0.371$, s.e. $= 0.167$; significant at the 5% level) and are 4.5km further from the dwelling ($\beta = 4.540$, s.e. $= 2.452$; significant at the 10% level). In terms of soil characteristics, though not statistically significant, plot soil is directionally of lower quality ($\beta = -0.076$, s.e. $= 0.080$), less likely to be "good" quality ($\beta = -0.021$, s.e. $= 0.061$), and less likely to be loam ($\beta = -0.105$, s.e. $= 0.080$).[14]

[14]Loam soils generally are more well-balanced and contain more nutrients and moisture than other soils.

Table 6: Which owned plots do households give out? Within-household comparison

	Fallow	Cultiv.	Log value	Distance	Soil qual.	Good soil	Loam
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Given-out plot (=1)	0.1839***	-0.1855**	-0.3708**	4.5403*	-0.0759	-0.0212	-0.1048
	(0.0646)	(0.0854)	(0.1667)	(2.4524)	(0.0800)	(0.0614)	(0.0798)
Mean of dep. var.	0.124	0.813	12.892	3.562	2.412	0.477	0.615
SD of dep. var.	0.329	0.390	1.502	13.294	0.611	0.500	0.487
Wave fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
HH fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,347	4,347	8,838	8,844	7,228	7,228	7,228
Households	1,006	1,006	1,616	1,617	1,471	1,471	1,471
Clusters	235	235	253	253	248	248	248
Within R^2	0.006	0.003	0.011	0.004	0.006	0.004	0.003

Notes: The unit of observation is a plot-year. The sample is the plot-year panel from the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13), restricted to household-lineages that are rural in every wave and present in all waves, and further restricted to owner-side plots. Each column regresses a plot attribute on an indicator for the owned plot being given out to others (lent free of charge) this wave: column (1) the plot was fallow in the previous wave; column (2) the plot was cultivated in the previous wave; column (3) the log of the household's perceived real sale value of the plot (base-2010 TSH); column (4) the distance from the plot to the dwelling (km); column (5) self-reported soil quality on a three-point scale (1=bad, 2=average, 3=good); column (6) an indicator for the top soil-quality category (good); column (7) an indicator for loamy soil. All specifications absorb household-lineage fixed effects and include wave fixed effects and time-varying household controls for head sex, head age, and head education (highest grade completed), so coefficients compare given-out with retained plots within the same household-lineage. Households is the number of distinct lineages contributing to the estimate. Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The borrower-side results are presented in Table 7. Within a recipient's portfolio, free-use plots are generally higher-quality relative to their other land. Free-use plots have higher quality soil ($\beta = 0.087$, s.e. $= 0.030$; significant at the 1% level) and households are ≈ 7 percentage points more likely to explicitly designate the plot's soil as "good", the highest category in the dataset ($\beta = 0.068$, s.e. $= 0.026$; significant at the 1% level). Though not quite statistically significant at conventional levels ($p = 0.170$), the plot's soil is also ≈ 4 percentage points more likely to be loam ($\beta = 0.039$, s.e. $= 0.028$).

Table 7: Are borrowed (free-use) plots worse land? Within-household comparison

	Soil quality (1–3) (1)	Soil good (=1) (2)	Loam (=1) (3)
Free-use plot (=1)	0.0869***	0.0682***	0.0390
	(0.0295)	(0.0257)	(0.0284)
Mean of dep. var.	2.413	0.478	0.614
SD of dep. var.	0.610	0.500	0.487
Wave fixed effects	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes
HH fixed effects	Yes	Yes	Yes
Observations	8,171	8,171	8,171
Households	1,604	1,604	1,604
Clusters	255	255	255
Within R^2	0.007	0.006	0.002

Notes: The unit of observation is a plot-year. The sample is the plot-year panel from the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13), restricted to household-lineages that are rural in every wave, and further restricted to owned and free-use plots. Each column regresses a plot-quality outcome on an indicator for the plot being used free of charge (borrowed) rather than owned: column (1) self-reported soil quality on a three-point scale (1=bad, 2=average, 3=good); column (2) an indicator for good soil quality; column (3) an indicator for loam soil. All specifications absorb household-lineage fixed effects and include wave fixed effects and time-varying household controls for head sex, head age, and head education (highest grade completed), so coefficients compare borrowed with owned plots within the same household-lineage. Households is the number of distinct lineages contributing to the estimate. Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Overall, these results tell a consistent story. Land lenders give out the peripheral plots in their portfolios. This land is less likely to be cultivated, is deemed of lower value, is directionally of lower soil quality, and lies farther from the dwelling. However, when borrowed plots are compared to other parcels within the recipients' portfolio, they have higher quality, loamier soil. These plots then present a material improvement in their portfolio, not just in plot quantity but also in plot quality, even if they were relatively "cheap" for the lender to share.

Free-use land borrowing follows household vulnerability to the shock A related exercise is to examine how the land-sharing margin operates across households with different vulnerabilities to the shock. By construction, risk-sharing implies that during shocks, resources will flow from less to more vulnerable households. If land is the margin across

which risk is shared, then land is expected to flow from land-rich to land-poor households. Land-rich households are more likely to have peripheral, uncultivated, or otherwise "excess" plots in a given season.

To test this, I divide the sample into tertiles based on the number of plots each household owns when it first enters the panel. I then define land-poor households as those in the bottom tertile, and land-rich households as those in the top tertile. Finally, I test the same specification presented in Table 5 (based on Equation 1), this time comparing the land-poor and land-rich subsamples. The findings are presented in Table 8. Consistent with the theory, for land-poor households, drought shocks generate statistically significant increases in the share ($\beta = 0.027$, s.e. $= 0.009$; significant at the 1% level), number ($\beta = 0.054$, s.e. $= 0.024$; significant at the 5% level), and presence (binary indicator, $\beta = 0.035$, s.e. $= 0.015$; significant at the 5% level) of free-use plots. These coefficients are approximately twice the magnitude of those in the pooled Table 5. Importantly, no significant results exist for land-rich households – drought shocks do not lead to an increase in their holding of free-use plots. This result is robust to using the *total plot acres owned* instead of plot count as the metric that the land-poor and land-rich tertiles are built on (Appendix Table 19). Owned plot count and acreage *per household member* yield similar results (Appendix Table 20, Appendix Table 21, Appendix Table 22, Appendix Table 23).

Table 8: Is drought-induced free-use uptake redistributive? Land-poor vs land-rich households

	Land-poor			Land-rich		
	Share	Number	Any	Share	Number	Any
	(1)	(2)	(3)	(4)	(5)	(6)
Drought shock (prior)	0.0273***	0.0544**	0.0347**	0.0049	0.0258	0.0104
	(0.0093)	(0.0242)	(0.0154)	(0.0050)	(0.0248)	(0.0142)
Mean of dep. var.	0.190	0.389	0.268	0.035	0.134	0.086
SD of dep. var.	0.353	0.793	0.443	0.137	0.515	0.281
Wave fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes	Yes	Yes
HH fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,193	1,193	1,229	1,079	1,079	1,081
Households	436	436	443	387	387	387
Clusters	174	174	175	162	162	162
Within R^2	0.025	0.026	0.021	0.044	0.040	0.037

Notes: The unit of observation is a household-year (the lineage-collapsed plot panel collapsed to one record per household-lineage and wave). The sample is the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13), restricted to household-lineages that are rural in every wave and present in all waves. Land-poverty is defined by tertiles of the household's number of owned plots at its first observed wave: columns (1)–(3) restrict to the bottom (land-poor) tertile and columns (4)–(6) to the top (land-rich) tertile. Within each group the outcome is the share of plots used free of charge (Share), the number of free-use plots (Number), and an indicator for holding any free-use plot (Any). **Drought shock (prior)** is the six-month Standardized Precipitation Index (SPI-6) for May of the prior year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. All specifications absorb household-lineage fixed effects and include wave fixed effects and time-varying household controls for head sex, head age, and head education (highest grade completed). Households is the number of distinct lineages contributing to the estimate. Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In Table 9, I ask whether the results in Table 5 are *driven* by the land-poor segment with an interacted model. I find that this is indeed the case; though drought shocks alone are not associated with a statistically significant increase in free-use plots at the household-level, the interaction with a dummy for land-poor households is positive and significant for the share ($\beta = 0.025$, s.e. $= 0.011$; significant at the 5% level), number ($\beta = 0.049$, s.e. $= 0.027$; significant at the 10% level) and presence ($\beta = 0.032$, s.e. $= 0.017$; significant at the 10% level) of free-use plots. The evidence provided here supports the hypothesis that following growing season drought shocks, free-use plots indeed flow to the most vulnerable households.

Table 9: Drought and free-use uptake: interaction with land-poverty

	Any free-use (1)	Share free-use (2)	No. free-use (3)
Drought shock (prior)	0.0042	0.0028	0.0077
	(0.0094)	(0.0048)	(0.0154)
Drought shock × Land-poor	0.0323*	0.0249**	0.0485*
	(0.0173)	(0.0108)	(0.0271)
Mean of dep. var.	0.160	0.098	0.230
SD of dep. var.	0.366	0.258	0.634
Wave fixed effects	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes
HH fixed effects	Yes	Yes	Yes
Observations	3,292	3,248	3,248
Households	1,183	1,174	1,174
Clusters	234	233	233
Within R^2	0.009	0.009	0.006

Notes: The unit of observation is a household-year (the lineage-collapsed plot panel collapsed to one record per household-lineage and wave). The sample is the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13), restricted to household-lineages that are rural in every wave. The outcome is an indicator for holding any free-use plot (column 1), the share of plots used free of charge (column 2), and the number of free-use plots (column 3). *Land-poor* equals one for households in the bottom tertile of owned plots at their first observed wave. **Drought shock (prior)** is the six-month Standardized Precipitation Index (SPI-6) for May of the prior year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. All specifications absorb household-lineage fixed effects and include wave fixed effects and time-varying household controls for head sex, head age, and head education (highest grade completed). Households is the number of distinct lineages contributing to the estimate. Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6 Discussion

In this paper, I show that land serves as a margin of insurance when rainfall drought, a covariate shock, strikes every household in a village at once and consumption-based risk-sharing collapses. Consistent with Townsend (1994) and Kazianga and Udry (2006), I demonstrate that drought shocks depress precisely the consumption-sharing channels that households would otherwise lean on, substantially reducing cash assistance received from other households and cutting the probability of relying on a friend or relative for food by about a third of its base rate. Against that backdrop, the gratis sharing of land moves in the opposite

direction: because a borrowed plot is productive only when cultivated, is not consumed by being lent, and ultimately reverts to its owner, gratis arrangements can hold or even expand at the moment the common pool of immediate consumption runs dry. My first test shows that borrowed plots become 14.3 percentage points less likely to be lost, a large effect against an 18.3% base rate. Reliance on free-use land expands on every margin: in share, count, and incidence. The second test follows from the observation that a plot is productive only to the extent it is cultivated, and that lending households hold peripheral parcels they cannot work in a given season. Lenders can better afford to share land if the plots given out are marginal in their portfolio rather than land the owner would otherwise farm. This is indeed the case. When compared with other plots in lender's portfolio, plots given out are roughly 19 percentage points less likely to have been cultivated the prior season, are further from the dwelling, and are of lower perceived value, so that lending costs the owner little foregone consumption. And, implied in any arrangement that shares risk, access flows toward need: the expansion in free-use land access is concentrated among land-poor households, for whom the estimates are about twice the pooled magnitude, with no greater uptake among the land-rich. These patterns hold across a battery of robustness tests.

Taken together, these findings contribute to each of the main literatures this thesis engages. To the risk-sharing canon, which has treated covariate shocks as uninsurable within the village and largely assumed them away (Kazianga and Udry, 2006; Townsend, 1994), the results offer a qualification: covariate risk defeats *consumption* sharing, as that literature holds and as I confirm in my sample, but it does not defeat sharing as such; a temporarily non-productive and non-depletable asset that reverts to its owner can be lent when there is no consumption left to give. To the literature measuring gains from marketizing land access (Acampora et al., 2025; Chen et al., 2023; de Janvry et al., 2015), in which non-marketed land, where it appears at all, enters as a residual, these results supply a new term absent in productivity accounting: the unpriced margin is doing insurance work while it persists, so the gains from marketization are, on this evidence, gross of an insurance value that has gone unmeasured.[15] To the largely qualitative literature on customary tenure, which has long asserted that African land rights are negotiated through membership and reciprocal obligation and that granted land serves a social-security function (Berry, 1993; Platteau, 1996, 2000), this paper supplies the empirical test that observation has lacked, and finds it holds: unpriced access expands when drought raises need and is drawn from the land lenders can most cheaply spare. Finally, relative to work on how households cope with shocks through migration, asset sales, or the rental market (Janzen and Carter, 2019; Kusunose and Lybbert, 2014; Morten, 2019; Tione and Holden, 2021), in which the landholder transacts over

[15] Munshi and Rosenzweig (2016) documents this tension between productive efficiency and risk-sharing within the context of migration. They find that individuals that benefit more from redistributive caste networks are less likely to migrate for work, thus leading to spatially inefficient labor allocation – a related dynamic.

land to absorb their own shock, I demonstrate a novel, unpriced mechanism running between households rather than within one.

Policy implications Here, I non-exhaustively explore land policy implications that follow from these findings.

First, because the informal insurance value of free-use land does not appear in productivity accounting, policies appraised on productivity grounds alone will systematically undervalue the unpriced margin and, in discouraging it, remove a form of drought support for vulnerable households. As a specific example, recall free-use lending often draws on non-cultivated land (Table 6) that owners either leave fallow, do not prioritize farming, or do not have the resources to work in a given season. Then, consider the incentives formed by how non-cultivation is treated in law and practice, particularly where customary tenure is strong. Goldstein and Udry (2008) show that Ghanaian farmers outside networks of social power are disproportionately likely to have land expropriated while it lies uncultivated.[16] An owner who fears the expropriation of a parcel they are unable to immediately farm has every reason to engage the priced margin (rent it, sell it) to avoid losing their plot. My findings indicate that this pressure will reduce the pool free-use plots draw from, dampening the effect of land-based risk-sharing under covariate shocks. Therefore, where customary land rights are strong and expropriation occurs, protecting non-cultivation is sharply relevant.

A second policy implication draws on the observation that a lender's fear that a loan becomes an ownership claim may constrain the supply of lent land. Though not studied on the free-use margin, evidence supports the concern that ambiguous ownership rights may limit land transactions: in Tigray, Holden et al. (2011) find that certifying the owner's title increased land supplied to the rental market by alleviating fears among lenders. A locally registrable instrument that records a gratis loan while documenting the owner's rights would address that fear explicitly by guaranteeing that the loan will not develop into an ownership claim. This prevents legal ambiguity around customary land-sharing from threatening its risk-sharing function.[17]

Third, the priced margin is not a substitute for the unpriced margin during shocks. Though rented land is often more productive (Acampora et al., 2025), particularly important during dry years with diminished harvests, and the rental market can provide a margin

[16] The statute in Tanzanian law is well tailored to address this issue. Tanzania's Village Land Act 1999 specifies allowing land to lie fallow among the purposes for which land may lawfully be occupied and used and sets a five-year threshold before abandonment can even be considered.

[17] This already exists in some contexts where customary tenure is strong. For example, Burkina Faso's rural land law provides dual registration: a land loan (*prêt de terre rurale*) may be entered in a register of rural land transactions, the law stipulates that recognized loans do not constitute acts of land possession, and the owner's rights are separately certified through an *attestation de possession foncière rurale* (Loi 034-2009, Arts. 36, 44–46, 52).

of adjustment during shocks (Eskander and Barbier, 2023; Kusunose and Lybbert, 2014; Tione and Holden, 2021), I find rented-in plots are 24.1 percentage points more likely to be surrendered under the same shock that protects free-use ones. Consequently, programs promoting land rental markets as a climate-adaptation instrument lean on a channel that can fail under drought shocks.

Fourth, land-sharing delivers land, but not necessarily the means to farm it. A household that receives such a plot after a bad year needs seed and other inputs at planting, months before harvest and at the trough of its liquidity. These households, being relatively land-poor, are also the least likely to be able to provide the large upfront investment that makes land productive. Policies, therefore, should encompass households farming land that they do not have formal stewardship of (rental agreement, title) and, further, should be timed to the post-shock planting window. If structured accordingly, they complement and reinforce the support that customary land-sharing provides.

Limitations and future research I now turn to discussing this paper's limitations and areas for expansion. A first limitation is that the NPS data does not link the two sides of a land loan. I observe, within lineage, that land-poor households take on more free-use land after drought, but I do not observe a plot passing from a specific lender to a specific borrower. Therefore, the compositional effects underlying my analysis are overall effects; still indicative, but they do not track specific transfers. A survey instrument that records loan counterparties and where both sides of each arrangement are enumerated would allow the reallocation to be observed directly and would permit a more specific test of the characteristics of the household portfolios and even prior reciprocity.

Second, my outcomes describe how land is allocated, not what that allocation achieves. I show that following drought shocks borrowed plots are retained and that reliance on them expands significantly among the land-poor, but I am unable to show that this measurably smooths consumption or buffers the negative welfare impact of the shock. Estimating this is not straightforward, since free-use uptake cannot be treated as exogenous in a consumption regression. The question of how much adaptive capacity informal land redistribution actually provides, and for how long, is the natural next step, and would likely require either an instrumental variable for access to free-use land or a setting with exogenous variation in the supply of it.

Third, three waves comprise only two plot transitions. As a consequence, the panel is too short to speak to long-run dynamics. I cannot say whether borrowed land is returned once the rains recover, which would make the arrangement a temporary reallocation, or whether it persists, which would make it a more durable redistribution; an event-study design over a longer panel would provide insight into long-run behavior after the shock.

Fourth, SPI enters linearly, which imposes a constant marginal effect and rules out the

threshold and nonlinear responses that Dell et al. (2014) notes. Broadly, different crops have different thresholds for "drought". Because crop water requirements and cropping calendars differ across Tanzania's agro-climatic zones, a measure calibrated to local crop mix would estimate the shock effect more precisely than a single standardized index.

Relatedly, my analysis focuses on the primary growing season in Tanzania. For much of the country, this is one unimodal season known as the *msimu* – my SPI-6 (measured December to May) shock covers almost all of the *msimu* (November to May).[18] However, in the North, Northeast, and some parts of the Northwest, there are two distinct growing (rainy) seasons: the long *masika* (March to May) and the short *vuli* (October to January). My shock measure covers the entirety of the *masika* but only part of the *vuli*. This is an intentional choice. First, in bimodal regions, the *masika* season is when primary crops (maize, sorghum, millet, rice) are grown. Second, *vuli* harvest contributes on average only 17 percent of national food crop production, while the main-season (*msimu* and *masika*) harvests captured by the December–May SPI-6 window account for over 80 percent (FAO and WFP, 1998). The *vuli* season is more volatile and, being shorter, is not the growing season many farmers rely on for the bulk of their production. This explanation aside, note that this bias is attenuating and these results are conservative. As with crop-specific drought measures, my insights can be strengthened by implementing these weather dynamics with further granularity.

Finally, the mechanism I document has a ceiling. I provide evidence that free-use lending operates on land-rich households' peripheral plots where that lending costs them little; that slack is what the insurance draws on. Under severe or sustained stress, the kind projected for Tanzania as dry spells lengthen (USAID, 2018), the slack may shrink as owners work more of their own land or as the shock exhausts the lending network, and the arrangement overstretches when it is most needed. Establishing where the ceiling lies is important for policy, since it determines whether customary land-sharing complements formal insurance or whether it fails alongside other risk-sharing as climate shocks intensify.

[18] The middle of the growing season coincides with the silking and flowering of key crops in the region. During this period, rainfall shocks experienced will have the greatest impact on yields at harvest (Barron et al., 2003; Denmead and Shaw, 1960).

References

Acampora, M., Casaburi, L., and Willis, J. (2025). Land rental markets: Experimental evidence from Kenya. *American Economic Review*, 115(3):727–771.

Barron, J., Rockström, J., Gichuki, F., and Hatibu, N. (2003). Dry spell analysis and maize yields for two semi-arid locations in east africa. *Agricultural and Forest Meteorology*, 117(1–2):23–37.

Beg, S. (2022). Digitization and development: Property rights security, and land and labor markets. *Journal of the European Economic Association*, 20(1):395–429.

Berry, S. S. (1993). *No Condition is Permanent: The Social Dynamics of Agrarian Change in Sub-Saharan Africa*. University of Wisconsin Press, Madison.

Chari, A. V., Liu, E. M., Wang, S.-Y., and Wang, Y. (2021). Property rights, land misallocation, and agricultural efficiency in China. *The Review of Economic Studies*, 88(4):1831–1862.

Chen, C. (2017). Untitled land, occupational choice, and agricultural productivity. *American Economic Journal: Macroeconomics*, 9(4):91–121.

Chen, C., Restuccia, D., and Santaeulàlia-Llopis, R. (2023). Land misallocation and productivity. *American Economic Journal: Macroeconomics*, 15(2):441–465.

Chimhowu, A. (2019). The 'new' African customary land tenure. characteristics, features and policy implications of a new paradigm. *Land Use Policy*, 81:897–903.

Cochrane, J. H. (1991). A simple test of consumption insurance. *Journal of Political Economy*, 99(5):957–976.

Colella, F., Lalive, R., Sakalli, S. O., and Thoenig, M. (2019). Inference with arbitrary clustering. IZA Discussion Paper 12584, IZA Institute of Labor Economics, Bonn.

Conley, T. G. (1999). GMM estimation with cross sectional dependence. *Journal of Econometrics*, 92(1):1–45.

de Janvry, A., Emerick, K., Gonzalez-Navarro, M., and Sadoulet, E. (2015). Delinking land rights from land use: Certification and migration in Mexico. *American Economic Review*, 105(10):3125–3149.

De Weerdt, J. and Dercon, S. (2006). Risk-sharing networks and insurance against illness. *Journal of Development Economics*, 81(2):337–356.

Dell, M., Jones, B. F., and Olken, B. A. (2014). What do we learn from the weather? the new climate-economy literature. *Journal of Economic Literature*, 52(3):740–798.

Denmead, O. T. and Shaw, R. H. (1960). The effects of soil moisture stress at different stages of growth on the development and yield of corn. *Agronomy Journal*, 52(5):272–274.

Eskander, S. M. S. U. and Barbier, E. B. (2023). Adaptation to natural disasters through the agricultural land rental market: Evidence from Bangladesh. *Land Economics*, 99(1):141–160.

Fafchamps, M. and Gubert, F. (2007). The formation of risk sharing networks. *Journal of Development Economics*, 83(2):326–350.

Fafchamps, M. and Lund, S. (2003). Risk-sharing networks in rural Philippines. *Journal of Development Economics*, 71(2):261–287.

FAO. Tanzania. Integrated Pest and Production Management (IPPM), FAO. Accessed: 2026-06-18.

FAO. Tanzania — family farming knowledge platform. FAO Family Farming Knowledge Platform. Accessed: 2026-06-18.

FAO and WFP (1998). FAO/WFP crop and food supply assessment mission to Tanzania. Special report, global information and early warning system on food and agriculture, Food and Agriculture Organization of the United Nations, Rome. 3 August 1998.

Fishman, R. (2011). Climate change, rainfall variability, and adaptation through irrigation: Evidence from indian agriculture. Job Market Paper, Columbia University.

Goldstein, M. and Udry, C. (2008). The profits of power: Land rights and agricultural investment in Ghana. *Journal of Political Economy*, 116(6):981–1022.

Gollin, D. and Udry, C. (2021). Heterogeneity, measurement error, and misallocation: Evidence from African agriculture. *Journal of Political Economy*, 129(1):1–80.

Gottlieb, C. and Grobovšek, J. (2019). Communal land and agricultural productivity. *Journal of Development Economics*, 138:135–152.

Holden, S. T., Deininger, K., and Ghebru, H. (2011). Tenure insecurity, gender, low-cost land certification and land rental market participation in Ethiopia. *The Journal of Development Studies*, 47(1):31–47.

Janzen, S. A. and Carter, M. R. (2019). After the drought: The impact of microinsurance on consumption smoothing and asset protection. *American Journal of Agricultural Economics*, 101(3):651–671.

Kazianga, H. and Udry, C. (2006). Consumption smoothing? livestock, insurance and drought in rural Burkina Faso. *Journal of Development Economics*, 79(2):413–446.

Kusunose, Y. and Lybbert, T. J. (2014). Coping with drought by adjusting land tenancy contracts: A model and evidence from rural Morocco. *World Development*, 61:114–126.

Lavigne Delville, P., Toulmin, C., Colin, J.-P., and Chauveau, J.-P. (2002). Negotiating access to land in West Africa: A synthesis of findings from research on derived rights to land. Technical report, International Institute for Environment and Development (IIED) and GRET, London.

Mace, B. J. (1991). Full insurance in the presence of aggregate uncertainty. *Journal of Political Economy*, 99(5):928–956.

Manysheva, K. (2022). Land property rights, financial frictions, and resource allocation in developing countries. STEG Working Paper 046, Structural Transformation and Economic Growth (STEG).

McKee, T. B., Doesken, N. J., and Kleist, J. (1993). The relationship of drought frequency and duration to time scales. In *Proceedings of the 8th Conference on Applied Climatology*, volume 17, pages 179–183, Boston, MA. American Meteorological Society.

Meghir, C., Mobarak, A. M., Mommaerts, C., and Morten, M. (2022). Migration and informal insurance: Evidence from a randomized controlled trial and a structural model. *Review of Economic Studies*, 89(1):452–480.

Miguel, E., Satyanath, S., and Sergenti, E. (2004). Economic shocks and civil conflict: An instrumental variables approach. *Journal of Political Economy*, 112(4):725–753.

Morten, M. (2019). Temporary migration and endogenous risk sharing in village India. *Journal of Political Economy*, 127(1):1–46.

Munshi, K. and Rosenzweig, M. (2016). Networks and misallocation: Insurance, migration, and the rural-urban wage gap. *American Economic Review*, 106(1):46–98.

Murken, L., Mager, G., Laudien, R., Kraehnert, K., and Gornott, C. (2024). The impact of weather risk on tenure security: Evidence from smallholder farmers in Tanzania. *Land Economics*, 100(4):668–689.

Platteau, J.-P. (1996). The evolutionary theory of land rights as applied to Sub-Saharan Africa: A critical assessment. *Development and Change*, 27(1):29–86.

Platteau, J.-P. (2000). *Institutions, Social Norms, and Economic Development*. Harwood Academic Publishers.

Tione, S. E. and Holden, S. T. (2021). Can rainfall shocks enhance access to rented land? evidence from Malawi. *Agricultural Economics*, 52(6):1013–1028.

Townsend, R. M. (1994). Risk and insurance in village India. *Econometrica*, 62(3):539–591.

Townsend, R. M. (1995). Consumption insurance: An evaluation of risk-bearing systems in low-income economies. *Journal of Economic Perspectives*, 9(3):83–102.

Udry, C. (1994). Risk and insurance in a rural credit market: An empirical investigation in northern Nigeria. *Review of Economic Studies*, 61(3):495–526.

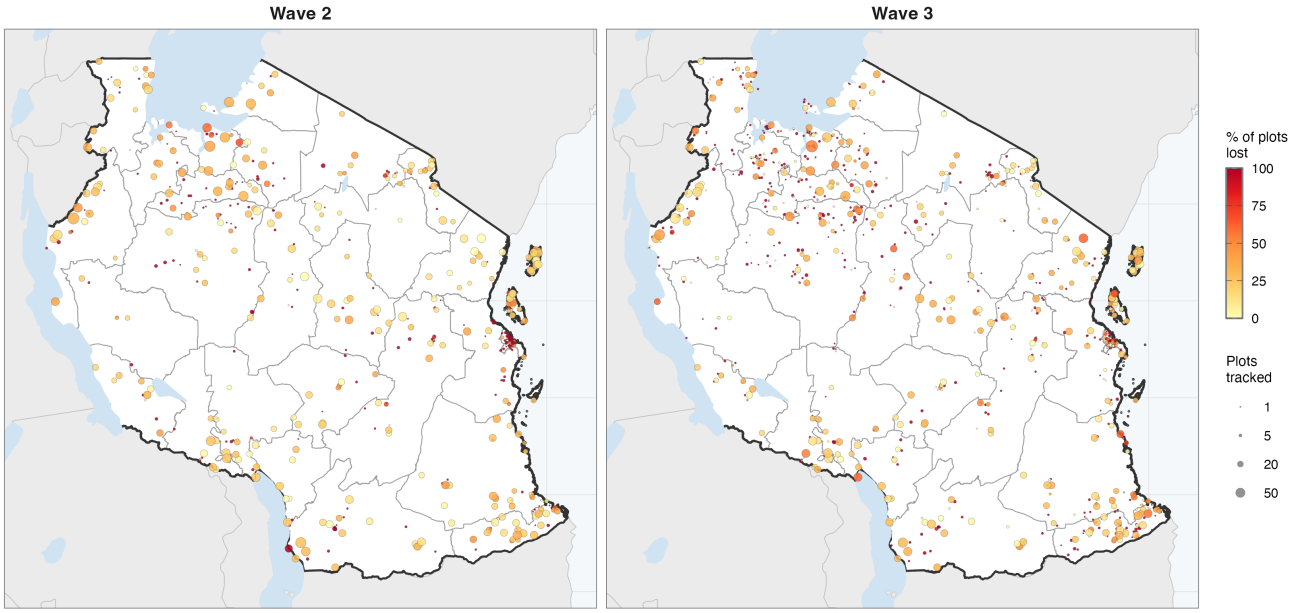
USAID (2018). Climate risk profile: Tanzania. Technical report, USAID.

Veljanoska, S. (2018). Can land fragmentation reduce the exposure of rural households to weather variability? *Ecological Economics*, 154:42–51.

Appendix

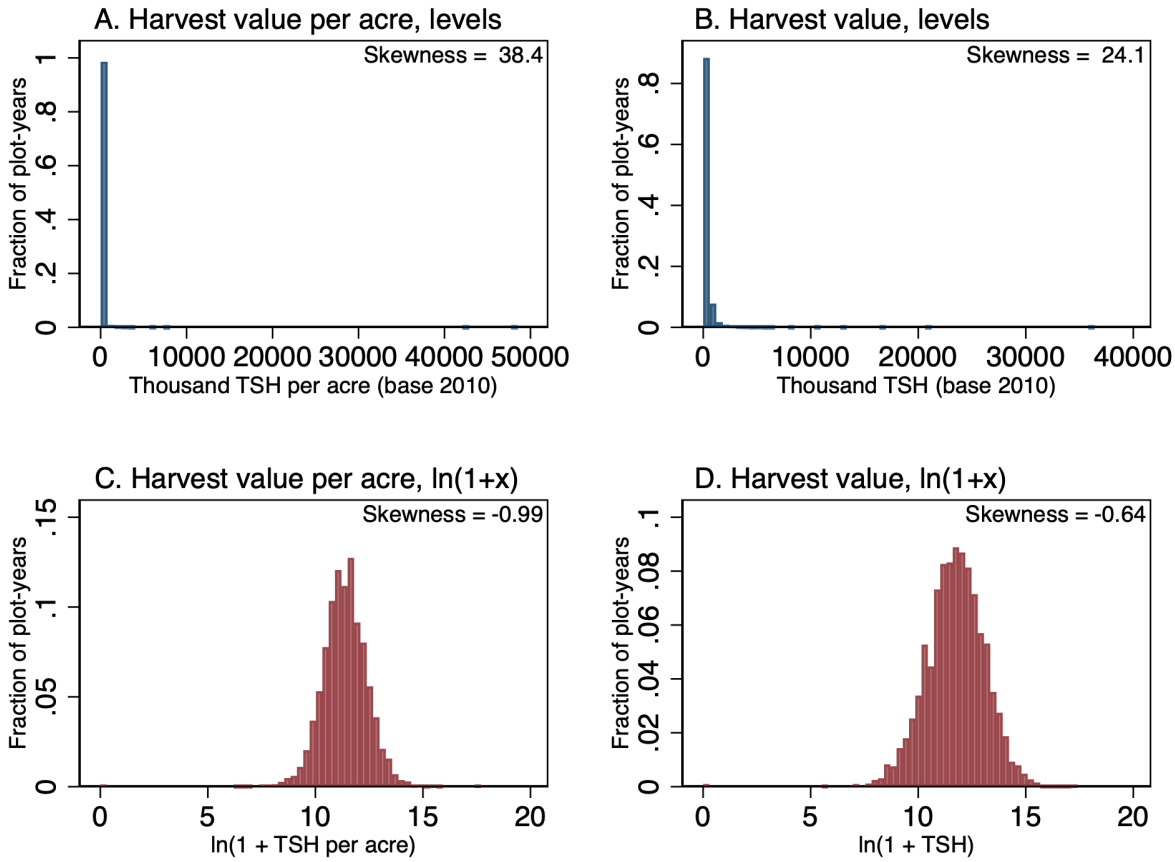
A Tables and Figures

Appendix Figure 1: Plot Loss Mapped, by Wave



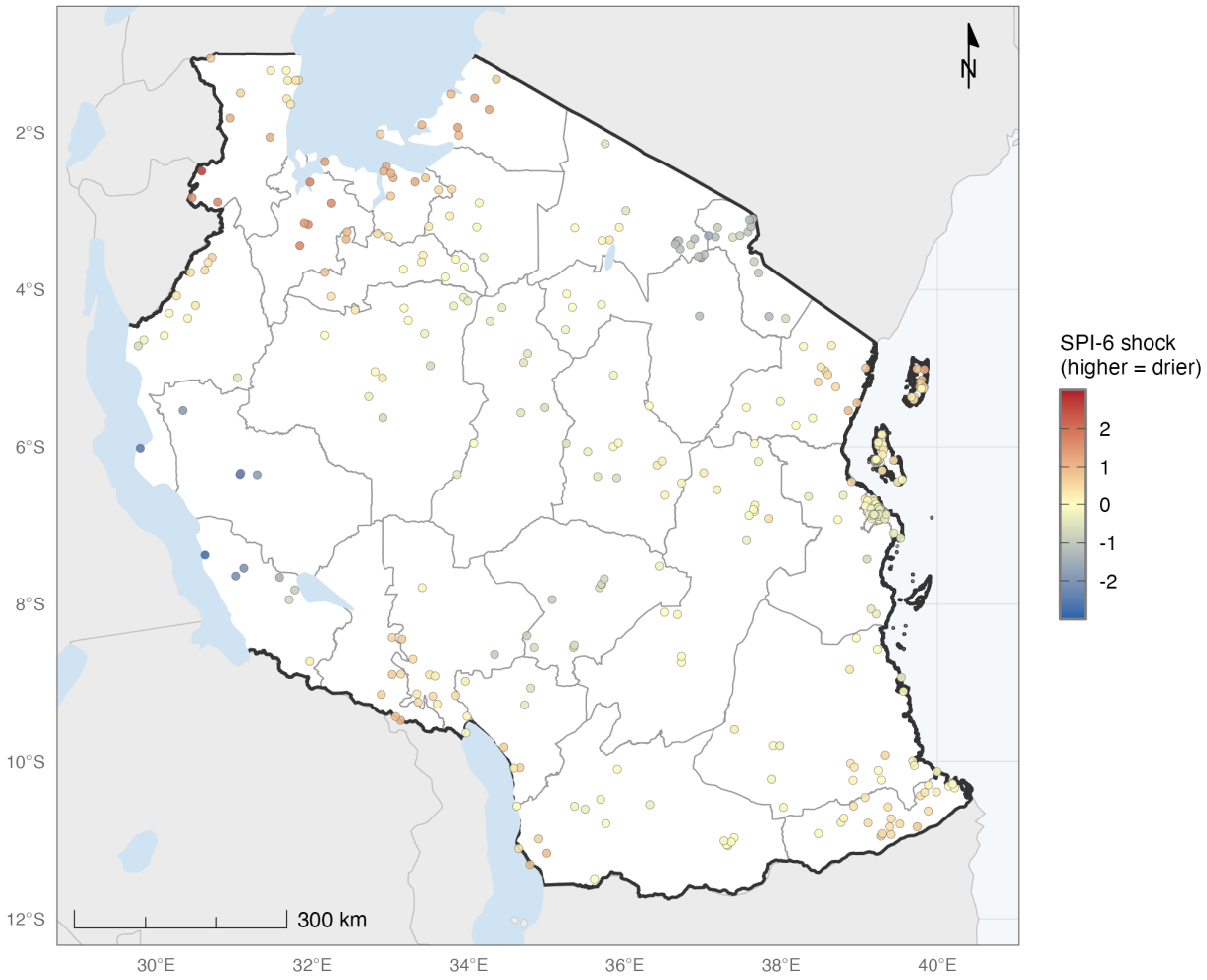
Notes: Plot loss by enumeration areas in Wave 2 (2010-2011) and Wave 3 (2012-2013). This is plot loss as compared with the prior wave across all tenure types.

Appendix Figure 2: Levels and Logs Histograms: Harvest value



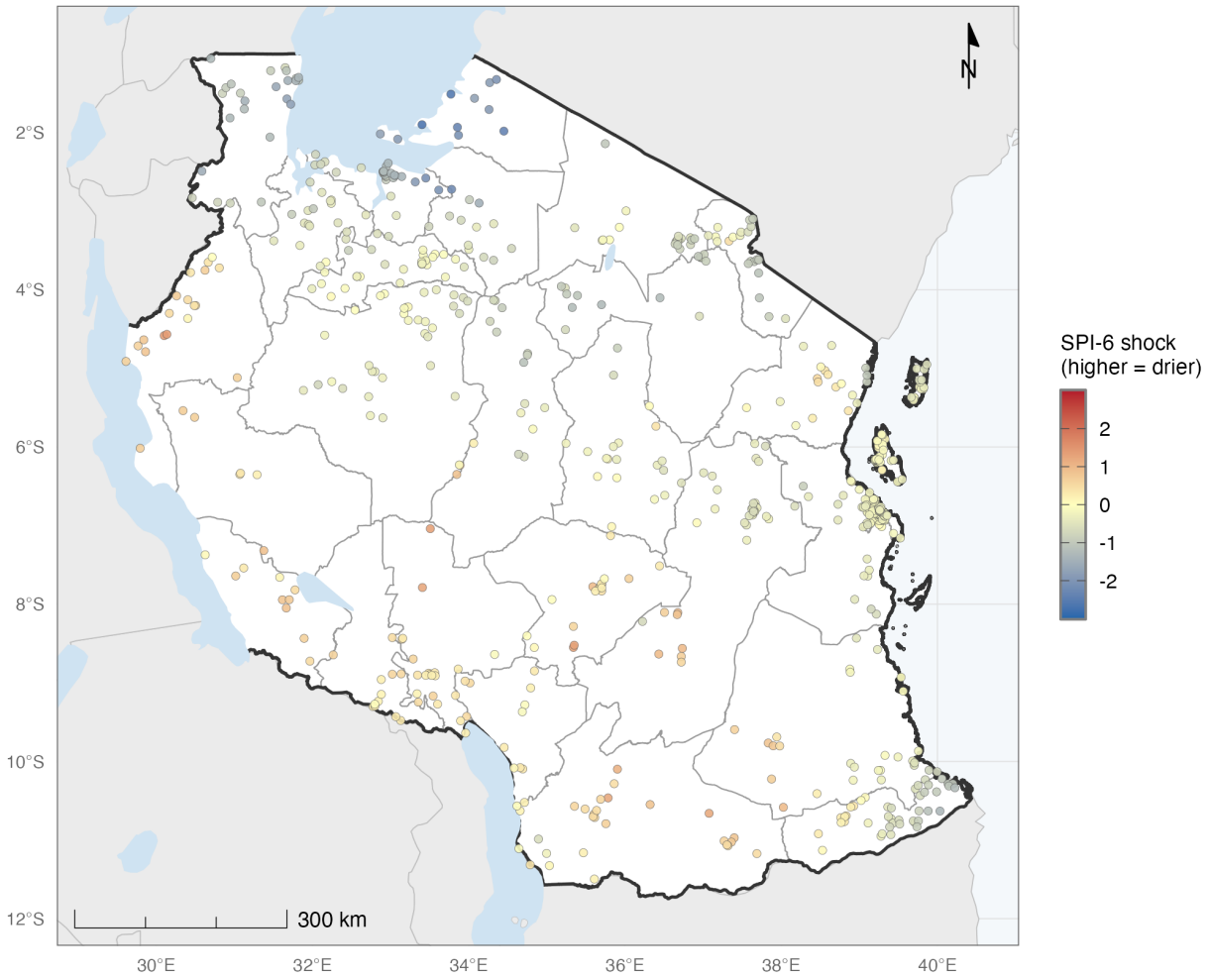
Notes: In levels, plot productivity statistics are heavily right-skewed with high skewness. Logs compress the upper tail and restore an approximately symmetric distribution.

Appendix Figure 3: SPI-6 Mapped, Wave 1



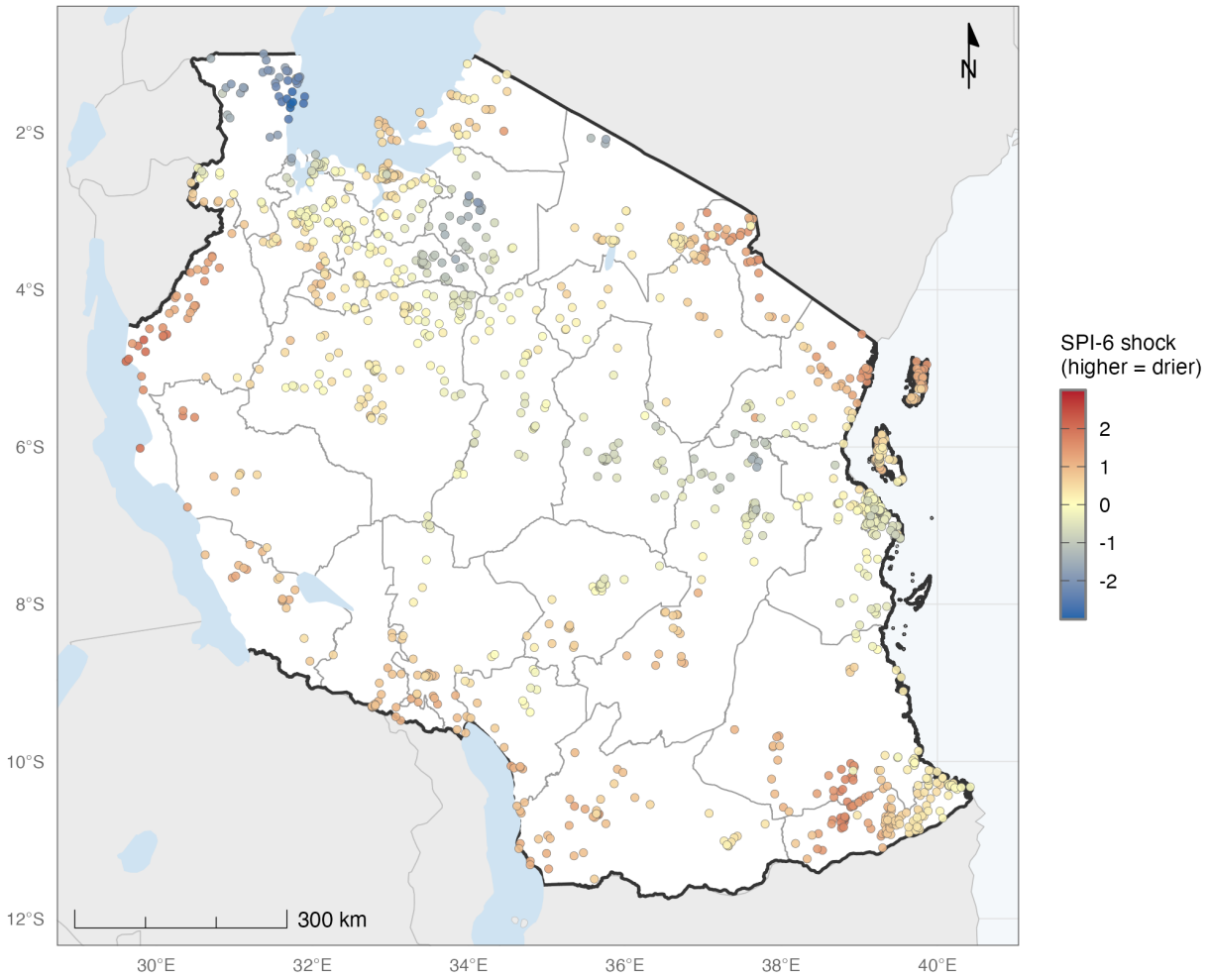
Notes: SPI-6 mapped to enumeration areas in Wave 1, 2008-2009. This figure comprises tracked survey locations. Note that SPI-6 is sign-flipped, such that higher is *drier* than the 1981–2007 norm.

Appendix Figure 4: SPI-6 Mapped, Wave 2



Notes: SPI-6 mapped to enumeration areas in Wave 2, 2010-2011. This figure comprises 714 tracked survey locations. Note that SPI-6 is sign-flipped, such that higher is *drier* than the 1981–2007 norm.

Appendix Figure 5: SPI-6 Mapped, Wave 3



Notes: SPI-6 mapped to enumeration areas in Wave 3, 2012-2013. This figure comprises 1,535 tracked survey locations. Note that SPI-6 is sign-flipped, such that higher is *drier* than the 1981–2007 norm.

Appendix Table 1: Balance of plot characteristics across tenure types

	Group means: mean (SD)			Pairwise differences			
	Owned (1)	Free-use (2)	Rented (3)	(2)−(1)	(3)−(1)	(3)−(2)	Joint p
Panel A. Size, location & agro-climate							
Plot area, self-report (acres)	2.956 (9.639)	1.255 (1.734)	1.723 (1.799)	-1.702***	-1.234***	0.468***	<0.001
Plot area, GPS (acres)	3.175 (7.483)	1.513 (2.835)	1.589 (1.527)	-1.663***	-1.586***	0.077	<0.001
Distance to dwelling (km)	5.101 (23.198)	6.814 (33.995)	12.598 (53.659)	1.713	7.498***	5.785**	<0.001
Rural location (=1)	0.869 (0.338)	0.829 (0.377)	0.815 (0.388)	-0.040***	-0.054***	-0.013	<0.001
Long-run rainfall (mm/yr)	829.4 (211.0)	783.9 (215.2)	816.8 (197.8)	-45.4***	-12.5	32.9***	<0.001
Panel B. Plot quality and physical attributes							
Soil quality good (=1)	0.471 (0.499)	0.439 (0.496)	0.497 (0.500)	-0.032**	0.026	0.059**	0.036
Erosion problem (=1)	0.123 (0.328)	0.099 (0.298)	0.117 (0.322)	-0.024***	-0.005	0.019	0.016
Steep slope (=1)	0.041 (0.198)	0.023 (0.151)	0.030 (0.170)	-0.017***	-0.011	0.006	0.001
Irrigated (=1)	0.019 (0.135)	0.017 (0.129)	0.065 (0.246)	-0.002	0.046***	0.048***	<0.001
Cultivated this season (=1)	0.798 (0.401)	0.885 (0.319)	0.868 (0.338)	0.087***	0.070***	-0.016	<0.001
Comfortable leaving fallow (=1)	0.951 (0.215)	0.623 (0.485)	0.256 (0.437)	-0.328***	-0.695***	-0.367***	<0.001
Panel C. Productivity and harvest (plots with a harvest record)							
Has harvest record (=1)	0.832 (0.374)	0.793 (0.405)	0.918 (0.274)	-0.039***	0.086***	0.125***	<0.001
Area harvested (acres)	2.564 (6.170)	1.512 (2.402)	2.030 (2.304)	-1.052***	-0.534***	0.518***	<0.001
Harvest value (000 real TSH)	261.3 (714.8)	174.6 (355.3)	312.8 (538.8)	-86.6***	51.5**	138.1***	<0.001
Harvest value p.a. (000 real TSH)	131.4 (176.2)	159.1 (195.3)	204.5 (259.7)	27.7***	73.1***	45.4***	<0.001
Log output per acre	11.22 (1.06)	11.43 (1.07)	11.63 (1.11)	0.21***	0.41***	0.20***	<0.001
Panel D. Household portfolio (holding context)							
Plots in HH portfolio	3.127 (1.703)	2.597 (1.329)	2.742 (1.738)	-0.530***	-0.385***	0.145	<0.001
Owned plots in HH	2.966 (1.705)	0.720 (1.016)	1.085 (1.293)	-2.247***	-1.881***	0.365***	<0.001
Free-use plots in HH	0.092 (0.364)	1.795 (0.982)	0.151 (0.451)	1.703***	0.059**	-1.644***	<0.001
Rented plots in HH	0.051 (0.277)	0.061 (0.346)	1.461 (0.832)	0.010	1.410***	1.399***	<0.001
Plot-year observations	17,798	2,220	850				
Lineages (clusters)	4,158	1,110	523				

Notes: The unit of observation is a plot-year, pooled across the three waves of the World Bank LSMS Tanzania National Panel Survey (NPS 2008/09, 2010/11, 2012/13) and restricted to plots present on the household's roster in a given wave; data are the lineage-collapsed plot panel, so each physical plot appears once per lineage. Plots are grouped by reported tenure: **Owned** (owned outright or shared ownership), **Free-use** (used free of charge / borrowed), and **Rented** (rented-in or sharecrop tenant). Columns (1)–(3) report group means with standard deviations in parentheses. Columns (4)–(6) report pairwise differences in means; stars denote the significance of each difference from a regression of the characteristic on tenure-group indicators with standard errors clustered at the lineage. The final column reports the p-value of a joint F-test that all three group means are equal. Plot areas, distance and harvest area are in their natural units, untrimmed and right-skewed; soil quality, slope, erosion and irrigation are self-reported (section 3A). *Comfortable leaving fallow* is a tenure-security proxy. Harvest value and output per acre are real, base-2010 TSH (output per acre winsorised at 1/99 within wave); these and the harvest measures in Panel C are defined only for plots with a harvest record, so their effective sample is smaller than the row counts at the foot of the table. Formal-title and sale-right indicators are omitted because they are recorded only for owner-side plots and therefore proxy tenure itself. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 2: Drought does not predict household or lineage attrition

	Household dropped (1)	Household-lineage dropped (2)
Drought shock	0.0010	-0.0102
	(0.0106)	(0.0318)
Observations	1,339	1,856
R^2	0.024	0.010

Notes: Linear probability models of panel attrition on baseline drought, with head controls. Column (1): among baseline (wave-1) rural households, an indicator for the household never being observed in wave 3. Column (2): among household-lineages rural in every observed wave, an indicator for the lineage not being tracked in every wave (and so excluded from the balanced plot sample). **Drought shock** is the six-month Standardized Precipitation Index in the unit's first observed wave, sign-flipped so higher values denote drier conditions; a positive coefficient would mean drought raises attrition. Neither estimate is significant. Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 3: Attrition and baseline characteristics: EA-clustered mean differences

	Stayer	Attritor	Diff.	p
Panel A. Household frame (baseline, wave 1; attrited = not seen in W3)				
Head female	0.243	0.251	-0.008	0.810
Head age	46.780	55.076	-8.296***	0.000
Head education (grade code)	16.751	15.260	1.491***	0.000
Household size	5.442	4.643	0.799***	0.000
Number of plots	2.340	2.168	0.172*	0.087
Log consumption p.a.e.	13.017	13.055	-0.038	0.419
Share free-use *(outcome)*	0.107	0.114	-0.007	0.767
Number free-use *(outcome)*	0.246	0.180	0.066*	0.080
Share owned	0.836	0.836	0.000	0.996
Share rented-in	0.049	0.040	0.009	0.513
Drought shock	0.078	0.100	-0.023	0.706
Panel B. Plot/lineage frame (rural lineages; dropped = excluded from balanced sample)				
Number of plots	2.382	2.383	-0.002	0.971
Share owned	0.845	0.849	-0.005	0.691
Share free-use *(outcome-related)*	0.099	0.105	-0.006	0.576
Share rented-in	0.048	0.039	0.008	0.166
Drought shock	0.086	-0.004	0.090	0.244

Notes: Baseline (first-observed) means by attrition status; **Diff.** is stayer minus attritor (Panel A) or retained minus dropped (Panel B), and p is from a mean-difference regression with standard errors clustered at the enumeration area. *Panel A* (household-level margin): wave-1 rural households, attrited = not observed in wave 3. Attrition is associated with demographics (attritors have older, less-educated heads and smaller households), but the free-use share, consumption, the owned and rented shares, and the shock are all balanced; stayers hold marginally more plots and free-use plots ($p \approx 0.08$). *Panel B* (plot-level margin): lineages rural in every observed wave, dropped = excluded from the balanced plot sample. Retained and dropped lineages are balanced on every portfolio dimension and on the shock. Regardless, level differences are absorbed by the household/lineage fixed effects and time-varying head controls, and drought does not differentially push free-use-holding lineages out of observation (Appendix Table 4). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 4: The outcome trajectory and free-use holding do not predict attrition

	Attrited by W3 (household) (1)	Attrited at t+1 (plot lineage) (2)
Δ free-use share (W1$\rightarrow$W2)	-0.0049	
	(0.0189)	
Plot loss recorded at W2	0.0167	
	(0.0179)	
Drought shock (higher = drier)		-0.0113*
		(0.0058)
Drought shock $\times$ free-use share		-0.0164
		(0.0262)
Frame	HH	Plot
Observations	1,225	3,623
R^2	0.029	0.036

Notes: Linear probability models testing selection on the within-unit identifying variation. Column (1): among rural households observed in waves 1 and 2, an indicator for not being observed in wave 3, on the wave-1-to-2 change in free-use share and on plot loss recorded at wave 2 (head controls included). Column (2): among at-risk rural lineage-waves, the next-wave attrition hazard on the drought shock (SPI-6) interacted with the lineage's free-use share. A null Δ free-use share (col. 1) means the outcome trajectory does not predict dropout; a null drought$\times$free-use interaction (col. 2) means drought-driven exit is not concentrated among free-use holders. Standard errors clustered at the enumeration area. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 5: Drought shock and household compositional changes

	Head sex changed (1)	Head replaced (2)	Size declined (3)
Drought shock (higher = drier)	-0.0045	-0.0027	-0.0053
	(0.0029)	(0.0045)	(0.0091)
Mean of dep. var.	0.015	0.043	0.210
Wave fixed effects	Yes	Yes	Yes
Observations	3,499	3,499	3,499
R^2	0.001	0.001	0.005

Notes: The unit of observation is a household-year and each outcome is a wave-to-wave change within the household: (1) an indicator for a change in the sex of the household head, (2) an indicator for a head replacement—a sex change or an age change inconsistent with the same head ageing, and (3) an indicator for a decline in household size. The sample is rural households, matching the household-level analysis. Because the outcomes are first differences, a household fixed effect is degenerate; each column is therefore pooled with wave fixed effects. Standard errors are clustered at the enumeration area, the level at which the shock varies; enumeration-area fixed effects are not included because SPI-6 is constructed from the lineage's fixed baseline coordinates and so varies essentially at that level. **Drought shock** is the six-month Standardized Precipitation Index for May of the prior year, sign-flipped so that higher values denote drier conditions; a positive coefficient would mean drought raises turnover. None of the coefficients is statistically significant: drought predicts neither a change of head nor a decline in household size. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 6: Rainfall shocks and agricultural productivity (plot fixed effects)

	Log harvest value (p.a.) (1)	Log harvest value (2)	Log maize yield (3)
Drought shock (concurrent)	-0.1191***	-0.0842**	-0.0857*
	(0.0431)	(0.0394)	(0.0445)
Mean of dep. var.	11.388	11.706	5.603
SD of dep. var.	1.085	1.347	0.955
Wave fixed effects	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes
Plot fixed effects	Yes	Yes	Yes
Observations	2,224	2,228	1,507
Plots	890	892	610
Clusters	100	100	97
Within R^2	0.018	0.019	0.009

Notes: This table reproduces the productivity estimates with plot fixed effects in place of household-lineage fixed effects, so identification comes from within-plot variation over time. The unit of observation is a plot-year. The sample is the plot-year panel from the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13), restricted to household-lineages that are rural in every wave, and to plots interviewed after the concurrent long-rains harvest. Each column is a separate plot fixed-effects regression of a log productivity outcome on the rainfall shock: column (1) log real harvest value per acre (base-2010 TSH), column (2) log real harvest value (base-2010 TSH), and column (3) log maize yield. **Drought shock (concurrent)** is the six-month Standardized Precipitation Index for May of the survey year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. All specifications absorb plot fixed effects and include wave fixed effects and time-varying household controls for head sex, head age, and head education (highest grade completed). Plots is the number of distinct plots contributing to the estimate. Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 7: Rainfall shocks and community participation

	No. coops	Coop members	Funds raised (m TSH)	No. projects
	(1)	(2)	(3)	(4)
Drought shock (prior)	-1.5822**	-31.0051	-3.8921	-0.0925
	(0.7979)	(23.8523)	(3.3240)	(0.0914)
Mean of dep. var.	4.096	143.387	36.137	2.050
SD of dep. var.	4.798	175.256	74.866	1.596
Wave fixed effects	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes
HH fixed effects	Yes	Yes	Yes	Yes
Observations	1,859	1,859	3,989	3,989
Households	1,124	1,124	1,812	1,812
Clusters	195	195	292	292
Within R^2	0.088	0.019	0.005	0.131

Notes: The sample is rural households in the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13). Each column is a separate within-household fixed-effects regression on the rainfall shock: column (1) the number of cooperative groups in the village; column (2) the number of farmers participating in cooperatives; column (3) total money raised for community projects (millions of TSH); column (4) the number of community investment projects implemented. **Drought shock (prior)** is the six-month Standardized Precipitation Index for May of the prior year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. All specifications include household fixed effects, wave fixed effects, and time-varying household controls for head sex, head age, and head education (highest grade completed). Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 8: Rainfall shocks and loss of free-use plots, close connection

	Dependent variable: plot lost (=1)
	Free-use (close connection)
	(1)
Drought shock (prior)	-0.1614*
	(0.0831)
Mean of dep. var.	0.182
SD of dep. var.	0.387
Wave fixed effects	Yes
Demographic controls	Yes
HH fixed effects	Yes
Observations	231
Households	85
Clusters	62
Within R^2	0.033

Notes: The unit of observation is a plot-year. The dependent variable is an indicator equal to one if a plot held by the household-lineage in wave $t-1$ is absent from its portfolio in wave t. The sample is the plot-year panel from the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13), restricted to household-lineages that are rural in every wave. The sample is limited to free-use plots: a plot enters the estimation sample only if it was used free of charge (borrowed) in the previous wave. Further, that plot must have been borrowed from a close connection (kin, employer, friend, or neighbor). **Drought shock (prior)** is the six-month Standardized Precipitation Index (SPI-6) for May of the prior year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. The specification absorbs household-lineage fixed effects and includes wave fixed effects and time-varying household controls for head sex, head age, and head education (highest grade completed). Households is the number of distinct lineages contributing to the estimate. Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 9: Rainfall shocks and plot loss by prior tenure

	Dependent variable: plot lost (=1)			
	All plots	Owned	Rented	Free-use
	(1)	(2)	(3)	(4)
Drought shock (prior)	0.0154	0.0179	0.2415***	-0.1432**
	(0.0120)	(0.0117)	(0.0659)	(0.0572)
Mean of dep. var.	0.118	0.090	0.483	0.183
SD of dep. var.	0.323	0.286	0.501	0.387
Wave fixed effects	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes
HH fixed effects	Yes	Yes	Yes	Yes
Observations	5,606	4,774	145	378
Households	1,174	1,058	54	129
Clusters	234	229	47	84
Within R^2	0.005	0.006	0.092	0.023

Notes: The unit of observation is a plot-year. The dependent variable is an indicator equal to one if a plot held by the household-lineage in wave $t-1$ is absent from its portfolio in wave t. The sample is the plot-year panel from the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13), restricted to household-lineages that are rural in every wave. Column (1) uses all plots; columns (2)–(4) restrict to plots that in the previous wave were, respectively, owned outright, rented-in or sharecropped (tenant side), and used free of charge (borrowed). **Drought shock (prior)** is the six-month Standardized Precipitation Index (SPI-6) for May of the prior year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. All specifications absorb household-lineage fixed effects and include wave fixed effects and time-varying household controls for head sex, head age, and head education (highest grade completed). Households is the number of distinct lineages contributing to the estimate. Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 10: Rainfall shocks and plot loss, by prior tenure (balanced panel)

	Dependent variable: plot lost (=1)			
	All plots	Owned	Rented	Free-use
	(1)	(2)	(3)	(4)
Drought shock (prior)	0.0154	0.0179	0.2415***	-0.1432**
	(0.0120)	(0.0117)	(0.0660)	(0.0572)
Mean of dep. var.	0.118	0.089	0.476	0.183
SD of dep. var.	0.322	0.285	0.501	0.387
Wave fixed effects	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes
HH fixed effects	Yes	Yes	Yes	Yes
Observations	5,591	4,762	143	378
Households	1,167	1,052	53	129
Clusters	234	229	47	84
Within R^2	0.005	0.006	0.092	0.023

Notes: The unit of observation is a plot-year. The dependent variable is an indicator equal to one if a plot held by the household-lineage in wave $t-1$ is absent from its portfolio in wave t. The sample is the plot-year panel from the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13), restricted to household-lineages that are rural in every wave and present in all waves. Column (1) uses all plots; columns (2)–(4) restrict to plots that in the previous wave were, respectively, owned outright, rented-in or sharecropped (tenant side), and used free of charge (borrowed). **Drought shock (prior)** is the six-month Standardized Precipitation Index (SPI-6) for May of the prior year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. All specifications absorb household-lineage fixed effects and include wave fixed effects and time-varying household controls for head sex, head age, and head education (highest grade completed). Households is the number of distinct lineages contributing to the estimate. Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 11: Rainfall shocks and plot loss by prior tenure (wet-tail-censored drought shock)

	Dependent variable: plot lost (=1)			
	All plots	Owned	Rented	Free-use
	(1)	(2)	(3)	(4)
Drought shock (prior, censored)	0.0307**	0.0327**	0.2476**	-0.1136**
	(0.0150)	(0.0145)	(0.1028)	(0.0569)
Mean of dep. var.	0.118	0.090	0.483	0.183
SD of dep. var.	0.323	0.286	0.501	0.387
Wave fixed effects	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes
HH fixed effects	Yes	Yes	Yes	Yes
Observations	5,606	4,774	145	378
Households	1,174	1,058	54	129
Clusters	234	229	47	84
Within R^2	0.006	0.007	0.065	0.013

Notes: This table reproduces the plot-loss-by-tenure estimates using a wet-tail-censored version of the drought shock, addressing the concern that a linear SPI treats unusually wet seasons as linearly better outcomes even though such seasons can themselves be harmful (e.g. through flooding). The unit of observation is a plot-year. The dependent variable is an indicator equal to one if a plot held by the household-lineage in wave $t-1$ is absent from its portfolio in wave t. The sample is the plot-year panel from the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13), restricted to household-lineages that are rural in every wave. Column (1) uses all plots; columns (2)–(4) restrict to plots that in the previous wave were, respectively, owned outright, rented-in or sharecropped (tenant side), and used free of charge (borrowed). **Drought shock (prior, censored)** is the six-month Standardized Precipitation Index (SPI-6) for May of the prior year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions, and censored below at -1: seasons more than one standard deviation wetter than the local norm are assigned the value -1, so identification comes from variation on the normal-to-dry range of the distribution. All specifications absorb household-lineage fixed effects and include wave fixed effects and time-varying household controls for head sex, head age, and head education (highest grade completed). Households is the number of distinct lineages contributing to the estimate. Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 12: Rainfall shocks and plot loss by prior tenure (three-month drought-shock specification)

	Dependent variable: plot lost (=1)			
	All plots	Owned	Rented	Free-use
	(1)	(2)	(3)	(4)
Drought shock (3-mo, prior)	0.0184*	0.0172*	0.2238**	-0.1355***
	(0.0097)	(0.0097)	(0.0982)	(0.0396)
Mean of dep. var.	0.118	0.090	0.483	0.183
SD of dep. var.	0.323	0.286	0.501	0.387
Wave fixed effects	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes
HH fixed effects	Yes	Yes	Yes	Yes
Observations	5,606	4,774	145	378
Households	1,174	1,058	54	129
Clusters	234	229	47	84
Within R^2	0.005	0.006	0.077	0.034

Notes: This table reproduces the plot-loss-by-tenure estimates using the three-month Standardized Precipitation Index (SPI-3) in place of the six-month index. The unit of observation is a plot-year. The dependent variable is an indicator equal to one if a plot held by the household-lineage in wave $t-1$ is absent from its portfolio in wave t. The sample is the plot-year panel from the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13), restricted to household-lineages that are rural in every wave. Column (1) uses all plots; columns (2)–(4) restrict to plots that in the previous wave were, respectively, owned outright, rented-in or sharecropped (tenant side), and used free of charge (borrowed). **Drought shock (3-mo, prior)** is the three-month Standardized Precipitation Index (SPI-3) for May of the prior year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. All specifications absorb household-lineage fixed effects and include wave fixed effects and time-varying household controls for head sex, head age, and head education (highest grade completed). Households is the number of distinct lineages contributing to the estimate. Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 13: Rainfall shocks and plot loss by prior tenure (Conley spatial standard errors)

	Dependent variable: plot lost (=1)			
	All plots	Owned	Rented	Free-use
	(1)	(2)	(3)	(4)
Drought shock (prior)	0.0154	0.0179	0.2415	-0.1432
Conley SE (50 km)	(0.0129)	(0.0125)	(0.0585)***	(0.0561)**
Conley SE (100 km)	(0.0144)	(0.0138)	(0.0662)***	(0.0545)***
Conley SE (200 km)	(0.0152)	(0.0153)	(0.0700)***	(0.0647)**
Mean of dep. var.	0.118	0.090	0.483	0.183
Households	1,174	1,058	54	129
Observations	5,601	4,769	145	378

Notes: This table reproduces the plot-loss-by-tenure estimates with Conley spatial standard errors in place of clustered standard errors. The point estimate in each column is the household-lineage fixed-effects estimate; beneath it are spatial heteroskedasticity- and autocorrelation-consistent standard errors (Conley, 1999) computed at distance cutoffs of 50, 100, and 200 km, allowing a one-wave temporal lag for within-lineage serial correlation, using the `acreg` estimator of Colella et al. (2019). Significance stars on each standard-error row refer to that cutoff. The unit of observation is a plot-year; the dependent variable is an indicator equal to one if a plot held by the household-lineage in wave $t-1$ is absent in wave t. The sample is the rural plot panel. Column (1) uses all plots; columns (2)–(4) restrict to plots that in the previous wave were owned, rented-in or sharecropped (tenant side), and used free of charge. **Drought shock (prior)** is the six-month Standardized Precipitation Index for May of the prior year (CHIRPS), sign-flipped so higher values denote drier conditions. All specifications absorb household-lineage and wave fixed effects and include head controls. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 14: Rainfall shocks and household reliance on free-use land (balanced panel)

	Share free-use (1)	No. free-use (2)	Any free-use (3)
Drought shock (prior)	0.0114**	0.0258*	0.0165*
	(0.0047)	(0.0140)	(0.0088)
Mean of dep. var.	0.105	0.238	0.236
SD of dep. var.	0.269	0.636	0.425
Wave fixed effects	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes
HH fixed effects	Yes	Yes	Yes
Observations	3,476	3,476	3,832
Households	1,388	1,388	1,541
Clusters	275	275	303
Within R^2	0.006	0.005	0.004

Notes: The unit of observation is a household-year. The sample is rural households in the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13). This specification is limited to the balanced panel of households. Each column is a separate within-household fixed-effects regression of the column outcome on the rainfall shock. Outcomes are the share of the household's plots that are used free of charge (borrowed) in column (1), the number of free-use plots in column (2), and an indicator for holding any free-use plot in column (3). **Drought shock (prior)** is the six-month Standardized Precipitation Index (SPI-6) for May of the prior year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. All specifications include household fixed effects, wave fixed effects, and time-varying household controls for head sex, head age, and head education (highest grade completed). Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 15: Rainfall shocks and household free-use reliance (Conley spatial standard errors)

	Share free-use (1)	No. free-use (2)	Any free-use (3)
Drought shock (prior)	0.0121	0.0272	0.0179
Conley SE (50 km)	(0.0056)**	(0.0147)*	(0.0088)**
Conley SE (100 km)	(0.0059)**	(0.0165)*	(0.0089)**
Conley SE (200 km)	(0.0059)**	(0.0160)*	(0.0094)*
Mean of dep. var.	0.102	0.235	0.232
Households	1,178	1,178	1,323
Observations	3,255	3,255	3,604

Notes: This table reproduces the household free-use estimates with Conley spatial standard errors in place of clustered standard errors. The point estimate in each column is the household fixed-effects estimate; beneath it are spatial heteroskedasticity- and autocorrelation-consistent standard errors (Conley, 1999) computed at distance cutoffs of 50, 100, and 200 km, allowing a one-wave temporal lag for within-household serial correlation, using the `acreg` estimator of Colella et al. (2019). Significance stars on each standard-error row refer to that cutoff. The unit of observation is a household-year. Outcomes are the share of the household's plots used free of charge (1), the number of free-use plots (2), and an indicator for holding any free-use plot (3). The sample is rural households with non-missing enumeration-area coordinates. **Drought shock (prior)** is the six-month Standardized Precipitation Index for May of the prior year (CHIRPS), sign-flipped so higher values denote drier conditions. All specifications absorb household-lineage and wave fixed effects and include head controls. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 16: The free-use crowd-in is not an artifact of household splitting

	Share free-use			No. free-use		
	(1)	(2)	(3)	(4)	(5)	(6)
Drought shock (prior)	0.0125***	0.0134**	0.0102	0.0272**	0.0381*	0.0083
	(0.0048)	(0.0063)	(0.0076)	(0.0137)	(0.0214)	(0.0166)
Lineage	All	No split	Split	All	No split	Split
Mean of dep. var.	0.109	0.114	0.088	0.240	0.250	0.207
Observations	3,874	2,292	1,407	3,874	2,292	1,407
Households	1,748	969	604	1,748	969	604
Within R^2	0.006	0.006	0.017	0.005	0.007	0.013

Notes: Household fixed-effects regressions of free-use reliance on the rainfall shock, with wave fixed effects and head controls; standard errors clustered at the enumeration area. The outcome is the share of the household's plots used free of charge (columns 1–3) and the number of free-use plots (columns 4–6). Columns (1) and (4) use all rural households (the published specification); columns (2) and (5) restrict to households whose lineage never splits; columns (3) and (6) to households whose lineage splits. The crowd-in is at least as strong among non-splitting households, so it reflects continuing households borrowing more rather than drought generating new, land-poor households. **Drought shock (prior)** is sign-flipped so higher values denote drier conditions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 17: Rainfall shocks and free-use plot loss (interaction, pre-harvest)

	Dependent variable: plot lost (=1)	
	Drought shock	Drought shock (3-mo)
	(1)	(2)
Drought shock (prior)	0.0160	
	(0.0159)	
Drought shock × Free-use	-0.1062**	
	(0.0524)	
Drought shock (3-mo, prior)		0.0325**
		(0.0132)
Drought shock (3-mo) × Free-use		-0.1151***
		(0.0435)
Mean of dep. var.	0.116	0.116
SD of dep. var.	0.320	0.320
Wave fixed effects	Yes	Yes
Demographic controls	Yes	Yes
HH × tenure fixed effects	Yes	Yes
Observations	3,197	3,197
Households	702	702
Clusters	155	155
Within R^2	0.011	0.014

Notes: The unit of observation is a plot-year. The dependent variable is an indicator equal to one if a plot held by the household-lineage in wave $t-1$ is absent from its portfolio in wave t. The sample is the plot-year panel from the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13), restricted to household-lineages that are rural in every wave. The sample is limited to households interviewed prior to the beginning of the contemporaneous season's harvest period. Each column interacts the rainfall shock with an indicator for the plot having been used free of charge (borrowed) in the previous wave; column (1) uses the six-month index (SPI-6) and column (2) the three-month index (SPI-3). **Drought shock (prior)** and **Drought shock (3-mo, prior)** are the six- and three-month Standardized Precipitation Indices (SPI-6 and SPI-3) for May of the prior year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. All specifications absorb household-lineage-by-prior-tenure fixed effects, which subsume the free-use main effect, and include wave fixed effects and time-varying household controls for head sex, head age, and head education (highest grade completed). Households is the number of distinct lineages contributing to the estimate. Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 18: Rainfall shocks and plot loss by prior tenure (pre-harvest)

	Dependent variable: plot lost (=1)			
	All plots	Owned	Rented	Free-use
	(1)	(2)	(3)	(4)
Drought shock (prior)	0.0134	0.0176	0.1894***	-0.1310
	(0.0161)	(0.0168)	(0.0618)	(0.0785)
Mean of dep. var.	0.122	0.093	0.435	0.176
SD of dep. var.	0.328	0.290	0.499	0.382
Wave fixed effects	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes
HH fixed effects	Yes	Yes	Yes	Yes
Observations	3,315	2,740	69	278
Households	718	634	27	88
Clusters	155	151	25	53
Within R^2	0.008	0.013	0.262	0.018

Notes: This table reproduces the plot-loss-by-tenure estimates restricting the sample to households interviewed before the start of the concurrent long-rains harvest. The unit of observation is a plot-year. The dependent variable is an indicator equal to one if a plot held by the household-lineage in wave $t-1$ is absent from its portfolio in wave t. The sample is the plot-year panel from the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13), restricted to household-lineages that are rural in every wave. Column (1) uses all plots; columns (2)–(4) restrict to plots that in the previous wave were, respectively, owned outright, rented-in or sharecropped (tenant side), and used free of charge (borrowed). **Drought shock (prior)** is the six-month Standardized Precipitation Index (SPI-6) for May of the prior year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. All specifications absorb household-lineage fixed effects and include wave fixed effects and time-varying household controls for head sex, head age, and head education (highest grade completed). Households is the number of distinct lineages contributing to the estimate. Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 19: Is drought-induced free-use uptake redistributive? Land-poor vs land-rich households (owned acreage)

	Land-poor			Land-rich		
	Share	Number	Any	Share	Number	Any
	(1)	(2)	(3)	(4)	(5)	(6)
Drought shock (prior)	0.0439***	0.0723***	0.0591***	-0.0016	0.0076	0.0038
	(0.0126)	(0.0257)	(0.0194)	(0.0053)	(0.0197)	(0.0104)
Mean of dep. var.	0.243	0.525	0.358	0.025	0.081	0.058
SD of dep. var.	0.382	0.901	0.480	0.120	0.394	0.234
Wave fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes	Yes	Yes
HH fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	951	951	1,004	1,161	1,161	1,166
Households	358	358	367	416	416	417
Clusters	160	160	162	179	179	179
Within R^2	0.028	0.020	0.038	0.022	0.020	0.012

Notes: The unit of observation is a household-year (the lineage-collapsed plot panel collapsed to one record per household-lineage and wave). The sample is the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13), restricted to household-lineages that are rural in every wave. Land-poverty is defined by tertiles of the household's total owned acreage at its first observed wave, where plot areas are GPS-measured where available and farmer-reported otherwise, summed over owned plots on the wave roster: columns (1)–(3) restrict to the bottom (land-poor) tertile and columns (4)–(6) to the top (land-rich) tertile. Within each group the outcome is the share of plots used free of charge (Share), the number of free-use plots (Number), and an indicator for holding any free-use plot (Any). **Drought shock (prior)** is the six-month Standardized Precipitation Index (SPI-6) for May of the prior year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. All specifications absorb household-lineage fixed effects and include wave fixed effects and time-varying household controls for head sex, head age, and head education (highest grade completed). Households is the number of distinct lineages contributing to the estimate. Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 20: Is drought-induced free-use uptake redistributive? Land-poor vs land-rich households (owned acreage per capita)

	Land-poor			Land-rich		
	Share	Number	Any	Share	Number	Any
	(1)	(2)	(3)	(4)	(5)	(6)
Drought shock (prior)	0.0361***	0.0667**	0.0472***	0.0021	0.0133	0.0062
	(0.0105)	(0.0286)	(0.0178)	(0.0065)	(0.0211)	(0.0119)
Mean of dep. var.	0.223	0.494	0.333	0.028	0.086	0.063
SD of dep. var.	0.370	0.886	0.472	0.128	0.404	0.243
Wave fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes	Yes	Yes
HH fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,040	1,040	1,092	1,139	1,139	1,145
Households	388	388	397	407	407	408
Clusters	161	161	163	169	169	170
Within R^2	0.025	0.019	0.032	0.013	0.012	0.007

Notes: The unit of observation is a household-year (the lineage-collapsed plot panel collapsed to one record per household-lineage and wave). The sample is the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13), restricted to household-lineages that are rural in every wave. Land-poverty is defined by tertiles of the household's total owned acreage per household member at its first observed wave, where plot areas are GPS-measured where available and farmer-reported otherwise, summed over owned plots on the wave roster and divided by household size: columns (1)–(3) restrict to the bottom (land-poor) tertile and columns (4)–(6) to the top (land-rich) tertile. Within each group the outcome is the share of plots used free of charge (Share), the number of free-use plots (Number), and an indicator for holding any free-use plot (Any). **Drought shock (prior)** is the six-month Standardized Precipitation Index (SPI-6) for May of the prior year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. All specifications absorb household-lineage fixed effects and include wave fixed effects and time-varying household controls for head sex, head age, and head education (highest grade completed). Households is the number of distinct lineages contributing to the estimate. Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 21: Drought and free-use uptake: interaction with land-poverty (owned acreage per capita)

	Any free-use	Share free-use	No. free-use
	(1)	(2)	(3)
Drought shock (prior)	0.0011	0.0023	0.0142
	(0.0089)	(0.0046)	(0.0143)
Drought shock × Land-poor	0.0461**	0.0275**	0.0356
	(0.0193)	(0.0126)	(0.0331)
Mean of dep. var.	0.166	0.099	0.232
SD of dep. var.	0.372	0.260	0.638
Wave fixed effects	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes
HH fixed effects	Yes	Yes	Yes
Observations	3,348	3,280	3,280
Households	1,203	1,190	1,190
Clusters	235	233	233
Within R^2	0.010	0.010	0.005

Notes: The unit of observation is a household-year (the lineage-collapsed plot panel collapsed to one record per household-lineage and wave). The sample is the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13), restricted to household-lineages that are rural in every wave. The outcome is an indicator for holding any free-use plot (column 1), the share of plots used free of charge (column 2), and the number of free-use plots (column 3). *Land-poor* equals one for households in the bottom tertile of total owned acreage per household member at their first observed wave, where plot areas are GPS-measured where available and farmer-reported otherwise, summed over owned plots on the wave roster and divided by household size. **Drought shock (prior)** is the six-month Standardized Precipitation Index (SPI-6) for May of the prior year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. All specifications absorb household-lineage fixed effects and include wave fixed effects and time-varying household controls for head sex, head age, and head education (highest grade completed). Households is the number of distinct lineages contributing to the estimate. Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 22: Is drought-induced free-use uptake redistributive? Land-poor vs land-rich households (owned plots per capita)

	Land-poor			Land-rich		
	Share	Number	Any	Share	Number	Any
	(1)	(2)	(3)	(4)	(5)	(6)
Drought shock (prior)	0.0262***	0.0526**	0.0395**	0.0013	0.0031	-0.0069
	(0.0090)	(0.0258)	(0.0159)	(0.0083)	(0.0316)	(0.0196)
Mean of dep. var.	0.183	0.382	0.257	0.039	0.125	0.083
SD of dep. var.	0.349	0.798	0.437	0.154	0.499	0.276
Wave fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes	Yes	Yes
HH fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,139	1,139	1,169	893	893	895
Households	415	415	421	322	322	322
Clusters	177	177	178	145	145	145
Within R^2	0.027	0.021	0.026	0.033	0.033	0.021

Notes: The unit of observation is a household-year (the lineage-collapsed plot panel collapsed to one record per household-lineage and wave). The sample is the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13), restricted to household-lineages that are rural in every wave and present in all waves. Land-poverty is defined by tertiles of the household's number of owned plots per household member at its first observed wave: columns (1)–(3) restrict to the bottom (land-poor) tertile and columns (4)–(6) to the top (land-rich) tertile. Within each group the outcome is the share of plots used free of charge (Share), the number of free-use plots (Number), and an indicator for holding any free-use plot (Any). **Drought shock (prior)** is the six-month Standardized Precipitation Index for May of the prior year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. All specifications absorb household-lineage fixed effects and include wave fixed effects and time-varying household controls for head sex, head age, and head education (highest grade completed). Households is the number of distinct lineages contributing to the estimate. Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 23: Drought and free-use uptake: interaction with land-poverty (owned plots per capita)

	Any free-use	Share free-use	No. free-use
	(1)	(2)	(3)
Drought shock (prior)	0.0004	0.0031	0.0083
	(0.0100)	(0.0053)	(0.0146)
Drought shock × Land-poor	0.0418**	0.0244**	0.0476
	(0.0186)	(0.0115)	(0.0291)
Mean of dep. var.	0.160	0.098	0.230
SD of dep. var.	0.366	0.258	0.634
Wave fixed effects	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes
HH fixed effects	Yes	Yes	Yes
Observations	3,292	3,248	3,248
Households	1,183	1,174	1,174
Clusters	234	233	233
Within R^2	0.010	0.009	0.006

Notes: The unit of observation is a household-year (the lineage-collapsed plot panel collapsed to one record per household-lineage and wave). The sample is the World Bank LSMS Tanzania National Panel Survey (NPS) waves 1–3 (2008/09, 2010/11, 2012/13), restricted to household-lineages that are rural in every wave. The outcome is an indicator for holding any free-use plot (column 1), the share of plots used free of charge (column 2), and the number of free-use plots (column 3). *Land-poor* equals one for households in the bottom tertile of owned plots per household member at their first observed wave. **Drought shock (prior)** is the six-month Standardized Precipitation Index for May of the prior year (CHIRPS, 1981–2007 calibration), sign-flipped so that higher values denote drier, more drought-affected conditions. All specifications absorb household-lineage fixed effects and include wave fixed effects and time-varying household controls for head sex, head age, and head education (highest grade completed). Households is the number of distinct lineages contributing to the estimate. Standard errors clustered at the enumeration area in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.